Introduction to information theory

Computational modeling of language
Spring 2026

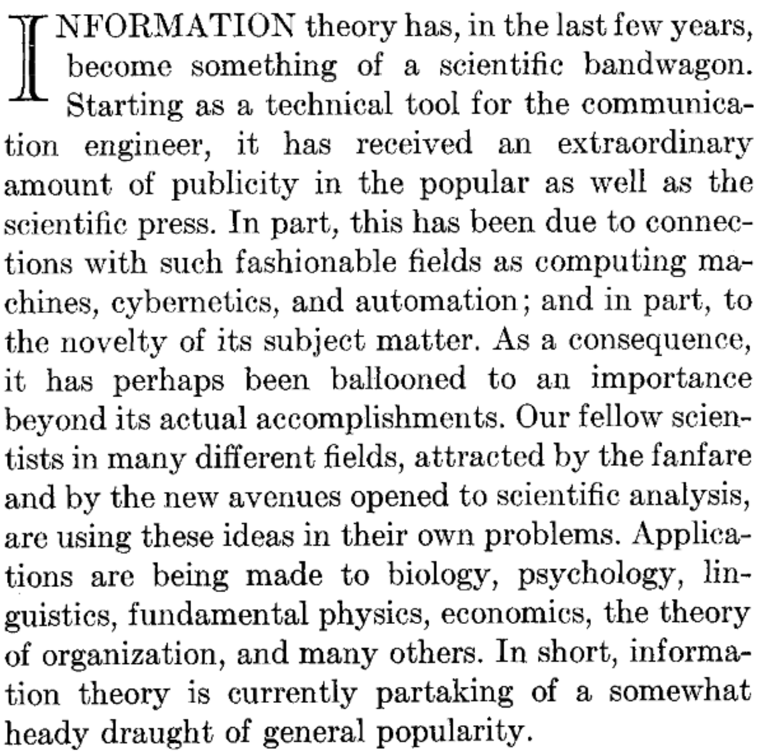
Bureaucracy

- Please supply anonymous mid-semester feedback:
  - https://tinyurl.com/3tmwbpe9

Our lineup for today

- Introduction to information theory
- Language and efficient communication
- Kinship

Shannon 1956


The Bandwagon

CLAUDE E. SHANNON

INFORMATION theory has, in the last few years, become something of a scientific bandwagon. Starting as a technical tool for the communication engineer, it has received an extraordinary amount of publicity in the popular as well as the scientific press. In part, this has been due to connections with such fashionable fields as computing machines, cybernetics, and automation; and in part, to the novelty of its subject matter. As a consequence, it has perhaps been ballooned to an importance beyond its actual accomplishments. Our fellow scientists in many different fields, attracted by the fanfare and by the new avenues opened to scientific analysis, are using these ideas in their own problems. Applications are being made to biology, psychology, linguistics, fundamental physics, economics, the theory of organization, and many others. In short, information theory is currently partaking of a somewhat heady draught of general popularity.

Based on reading by

Stone (2018)

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Claude Shannon, 1985

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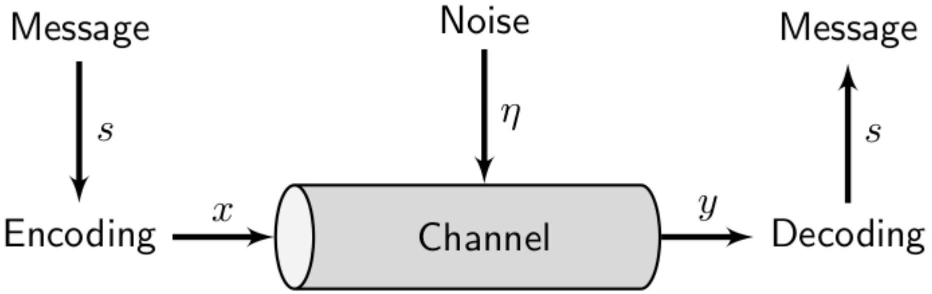

Figure 1: The communication channel. A message (data) is encoded before being used as input to a communication channel, which adds noise. The channel output is decoded by a receiver to recover the message.

Approach: Work from the ground up, define basic quantities first, then others that rely on them.

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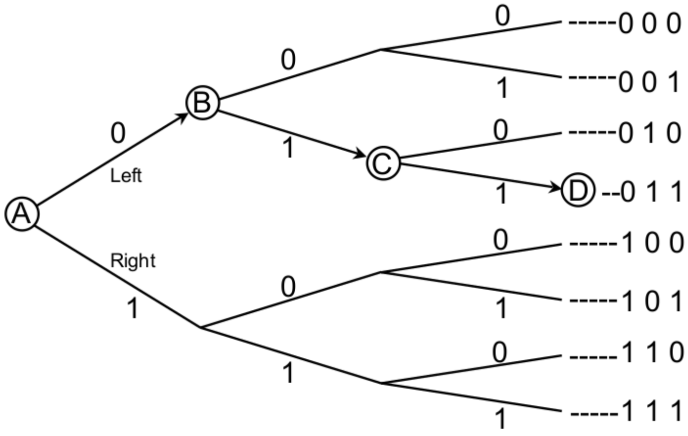
Bit: Unit of information, the amount of information you need to make one unbiased binary decision, e.g. a flip of a fair coin, a fork in the road.



Information as surprise. The Shannon information, or **surprisal**, of a particular outcome x_h (e.g. heads) is:

$$\log 1 / p(x_h) \text{ bits} \quad = \quad - \log p(x_h) \text{ bits}$$

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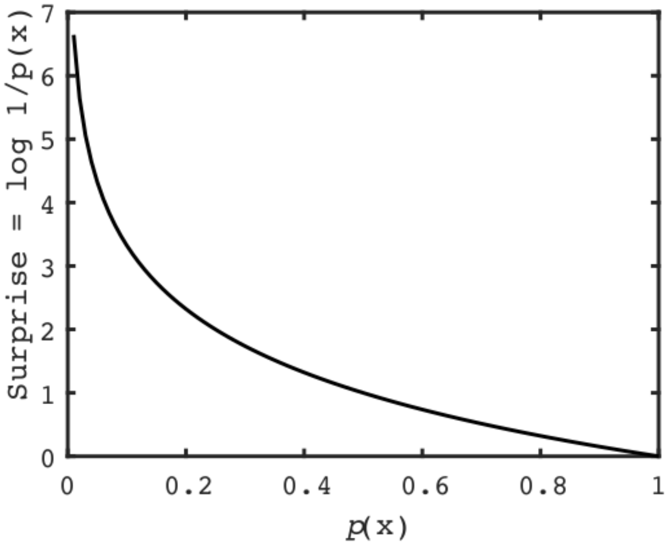

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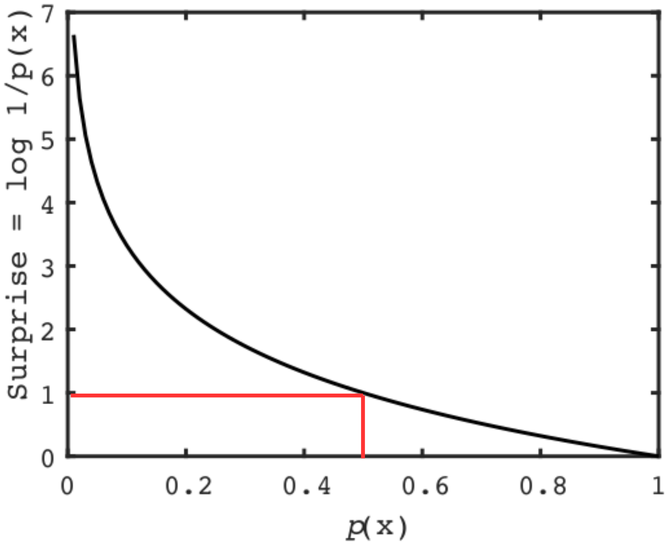

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A: The surprise you feel upon seeing an outcome is an index of **how much information you have just gained** by seeing that outcome.

[4]:

```
# sanity check
# find the US pronunciation of the word "entropy"
df[df.Word=='entropy']
```

[4]:

	Word	Pron
17502	entropy	[ɛ, n, t, ɹ, ə, p, i]

Entropy is <u>expected surprisal</u>, given by a weighted average over all possible outcomes.

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If we have a random variable *x* with probability distribution *p*(*x*), its entropy is:

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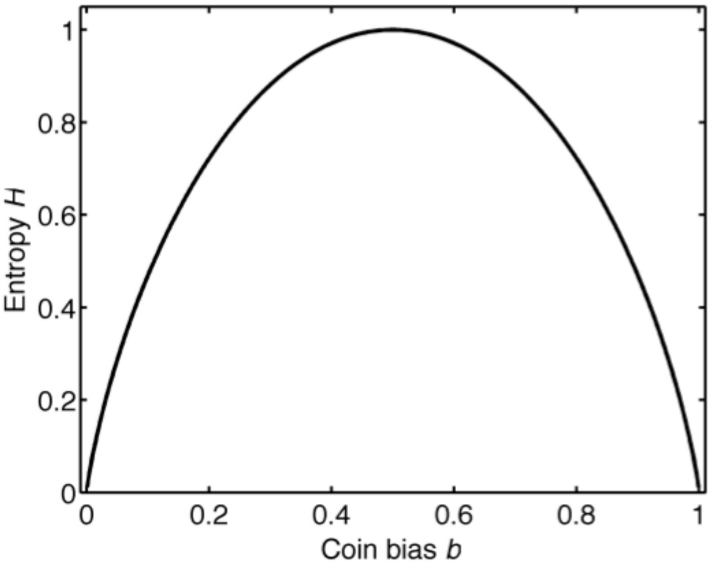
This tells you how much information **on average** you expect to gain by observing the outcome of variable *x*.

Entropy H shown as a function of coin bias, i.e. the probability $p(x_h)$ of a coin coming up heads.



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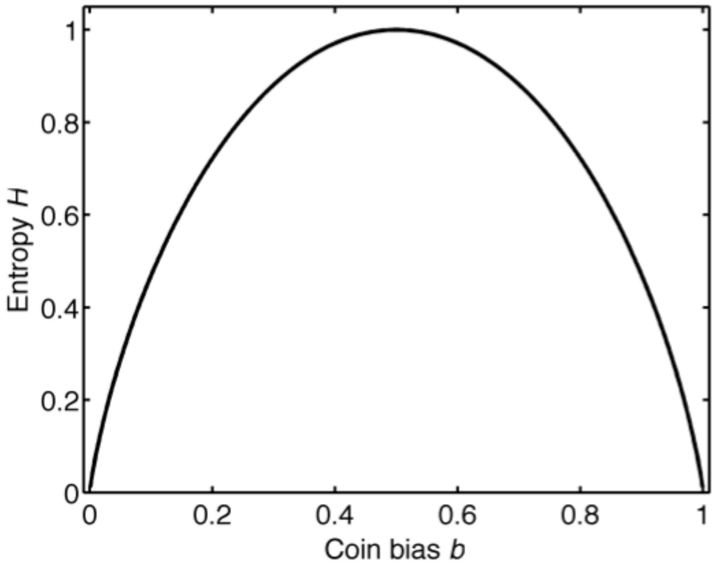

Entropy is maximal for a uniform distribution:
$p(x_h) = p(x_t) = 0.5$.

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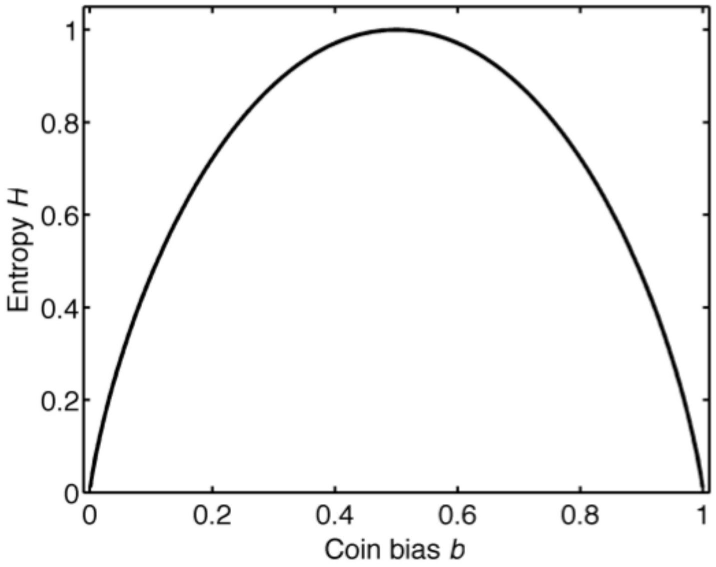

A measure of **uncertainty** about the outcome.

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Principle of maximum entropy

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- Must be **testable knowledge** that places **constraints** on the distribution, e.g.:

  “The mean value of variable *X* is 2.87.”

Principle of maximum entropy

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- Epistemic modesty
- Minimize assumptions
- Occam’s razor

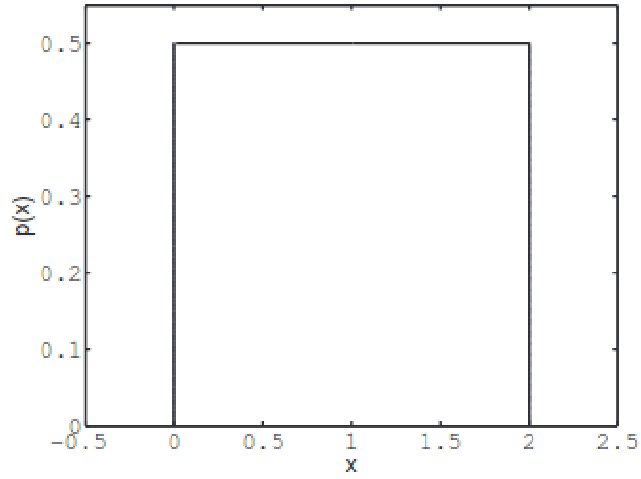
Principle of maximum entropy

Of all probability distributions that represent a given state of knowledge, the one that **best** represents that knowledge is the one with **maximum entropy**.

- Finding the maximum entropy (MaxEnt) distribution that satisfies a given set of constraints is an instance of **constrained optimization**.

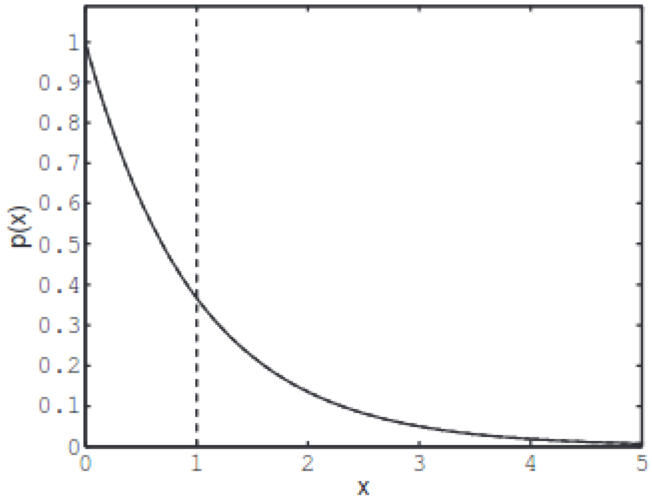
Maximum entropy distributions

Example 1: If a variable *x* has a fixed **lower bound** *xmin* and **upper bound** *xmax*, but is otherwise unconstrained, then the maximum entropy distribution is **uniform**:



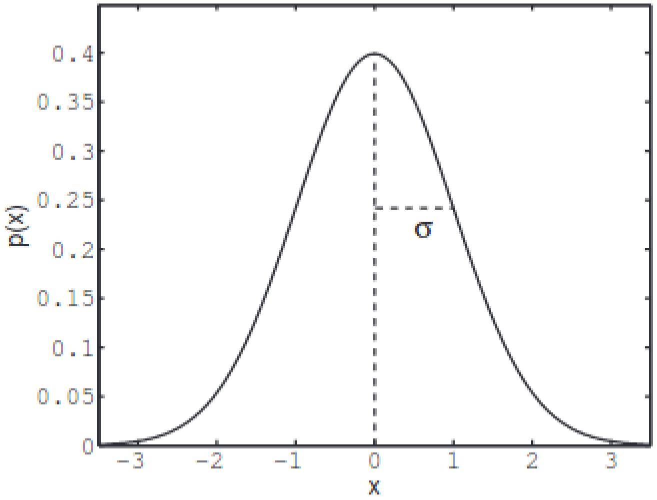
Example 2: If a variable *x* has **no values below 0**, and has a fixed **mean** *μ*, but is otherwise unconstrained, then the maximum entropy distribution is **exponential**:

$$p(x) = \frac{1}{\mu} e^{-x/\mu}$$



Example 3: If a variable *x* has a fixed **variance** but is otherwise unconstrained, then the maximum entropy distribution is **Gaussian, a.k.a. normal**:

$$p(x) \quad = \quad \frac{1}{\sqrt{2\pi v_x}} e^{-(\mu_x - x)^2/(2v_x)}$$



Thus, these well-known distributions are special cases of a more general principle - that of maximum entropy.

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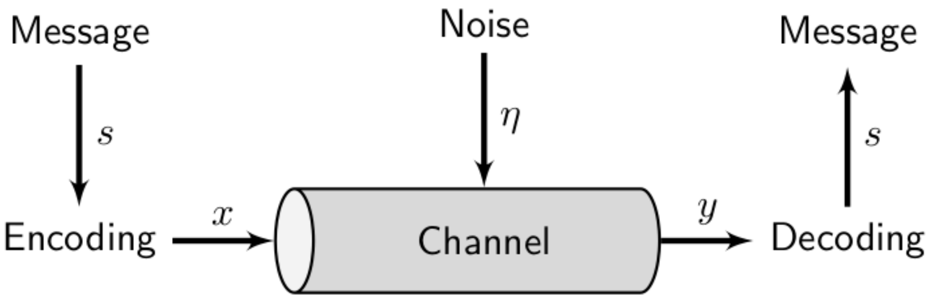

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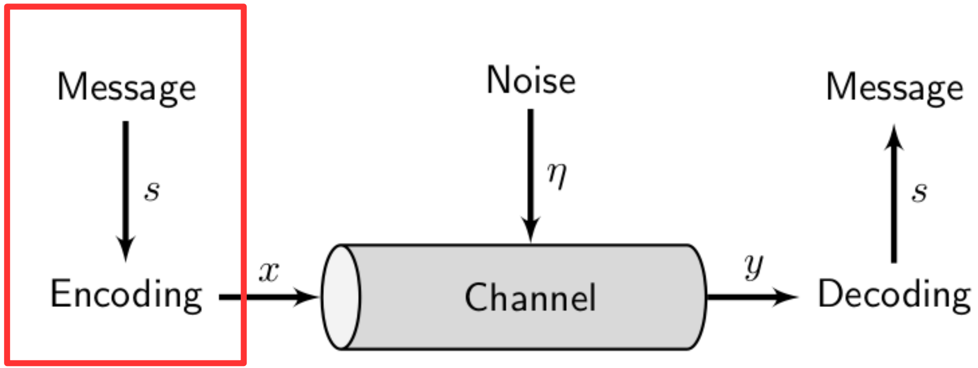

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If we are given a random variable *s* with probability distribution *p*(*s*), and with entropy *H*(*s*) as just defined, ...

… then each value of the variable *s* can be represented with an average of (just over) *H*(*s*) binary digits, with negligible risk of information loss.

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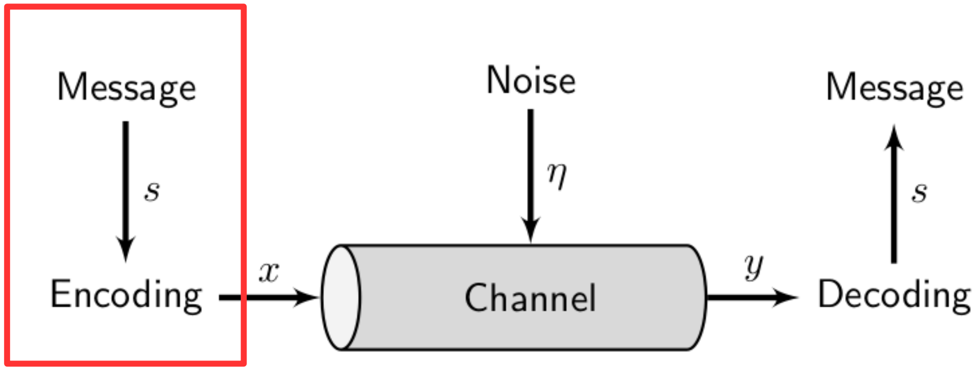

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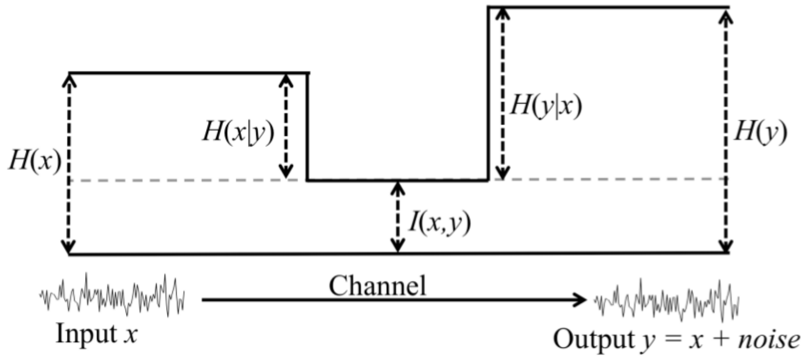

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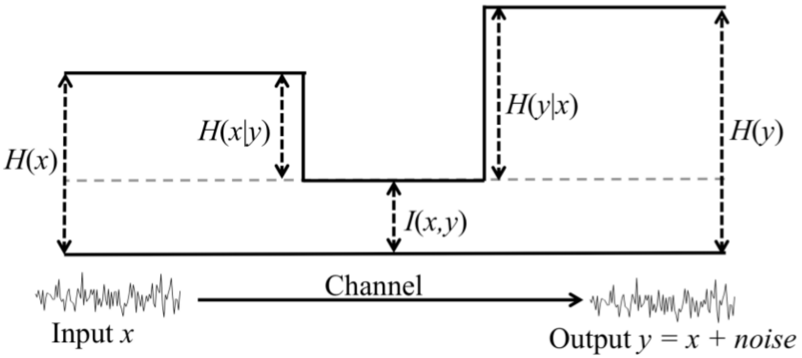
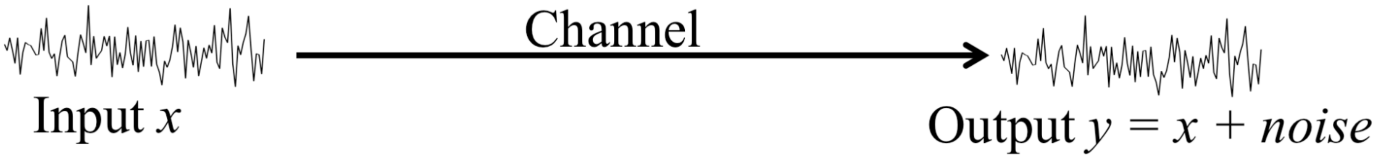
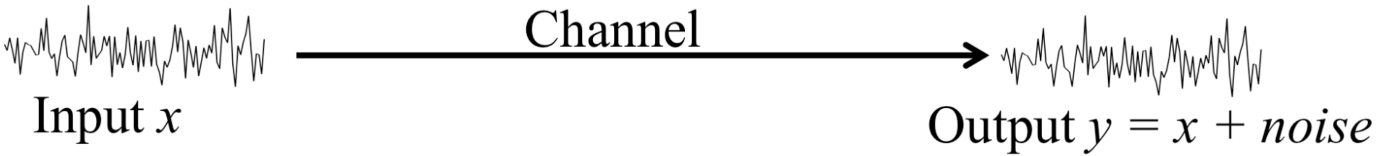


The **amount of information** we get about **one random variable**, by observing the value of **another**.


Channel
Input x
Output $y = x + noise$



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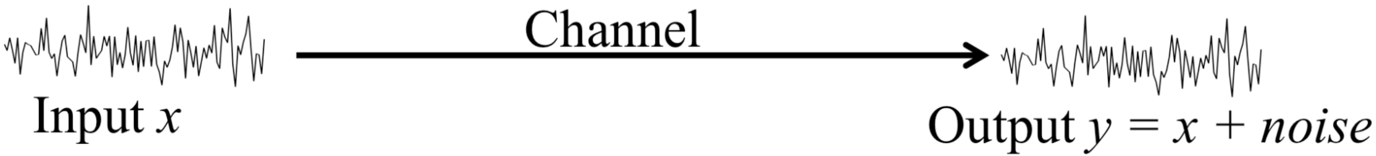

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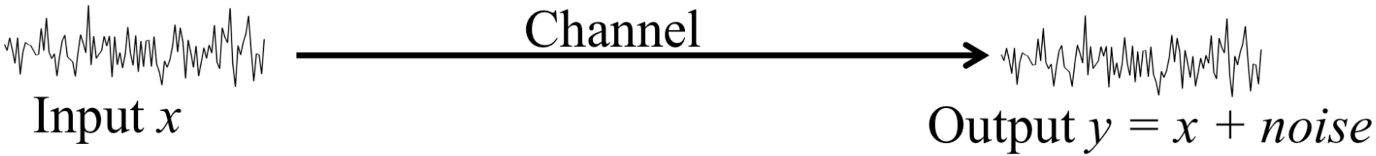

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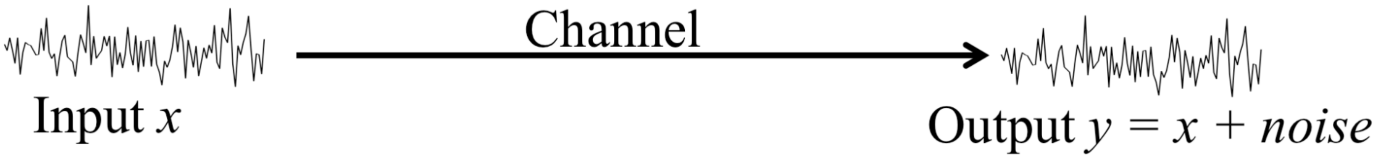

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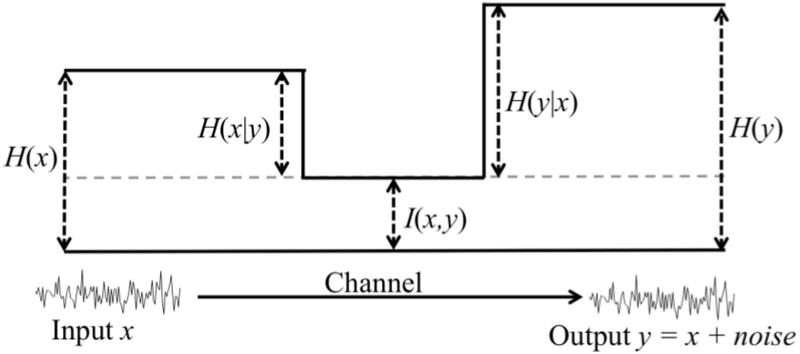

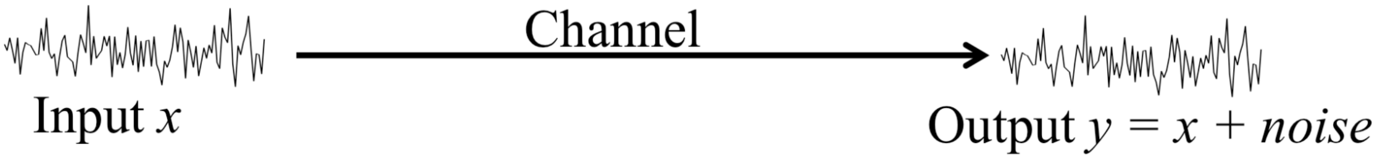

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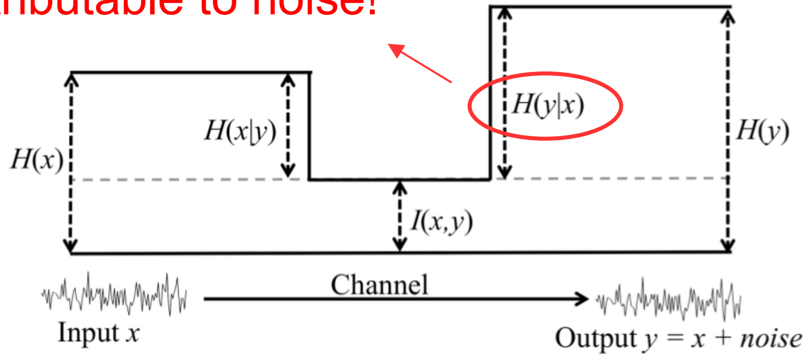
Attributable to noise!



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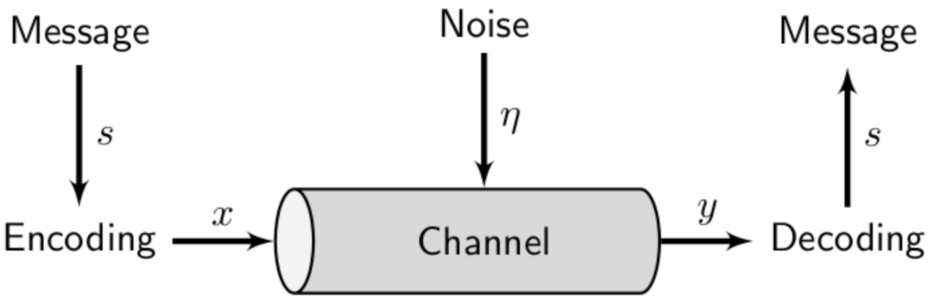

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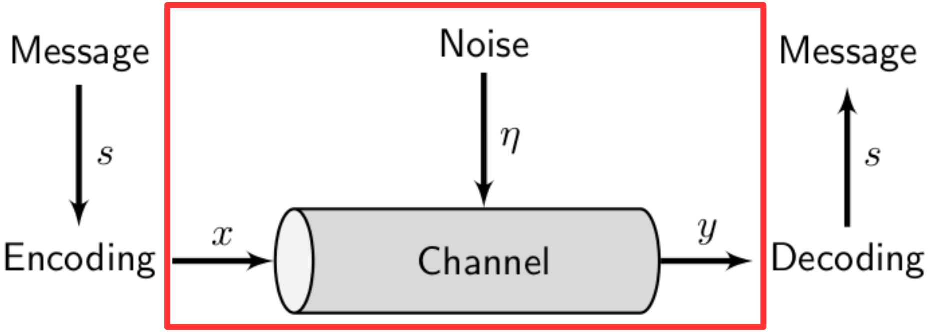

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[I]t is possible to use a communication channel to communicate information at a rate arbitrarily close to the channel capacity of *C* bits/s, with arbitrarily low frequency of errors.

But it is not possible to communicate information with arbitrarily low frequency of error at a rate greater than *C* bits/s.

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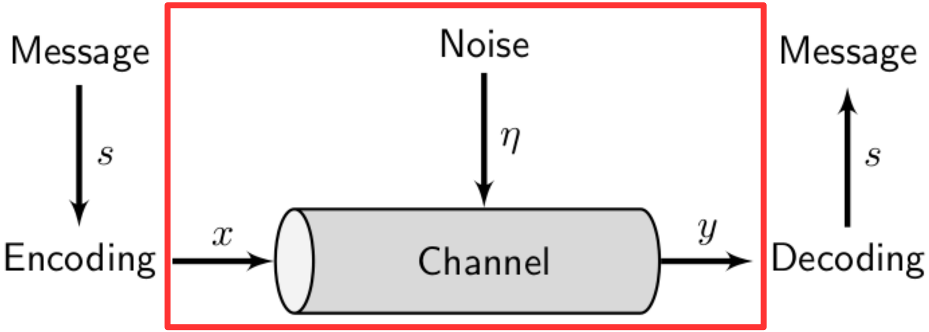

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Core concepts:

Bit, as unit of information
Surprisal, information as surprise
Entropy, or expected surprisal, as measure of uncertainty
Principle of maximum entropy
Source coding theorem
Mutual information between two variables
Channel capacity
Noisy channel theorem

What **questions** do you have?

What **comments** do you have?

Trends in Cognitive Sciences

Feature Review

How Efficiency Shapes Human Language

Edward Gibson,[1,*] Richard Futrell,[2] Steven T. Piandadosi,[3] Isabelle Dautriche,[4] Kyle Mahowald,[5] Leon Bergen,[5] and Roger Levy[1,*]

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CellPress REVIEWS

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- Background
- Novel contribution
- Evidential and/or theoretical basis for the contribution

Background (G et al.)

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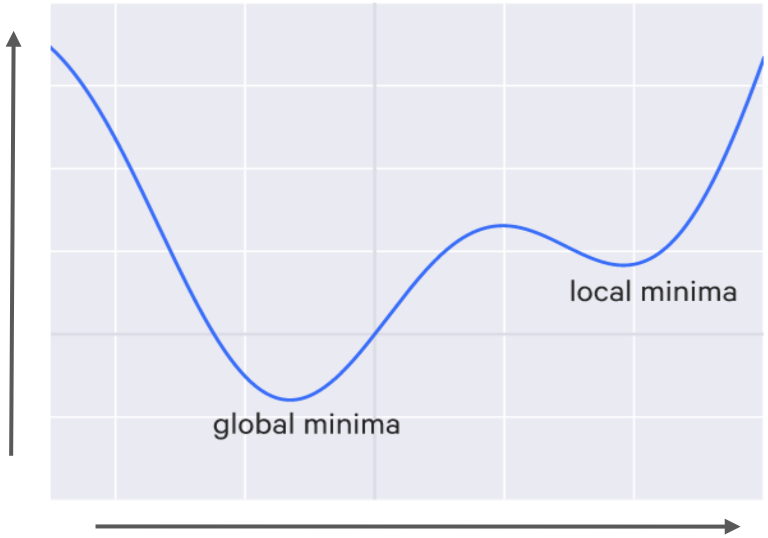
- Some have argued that human languages can be viewed as **local optima** in a landscape of possible languages.



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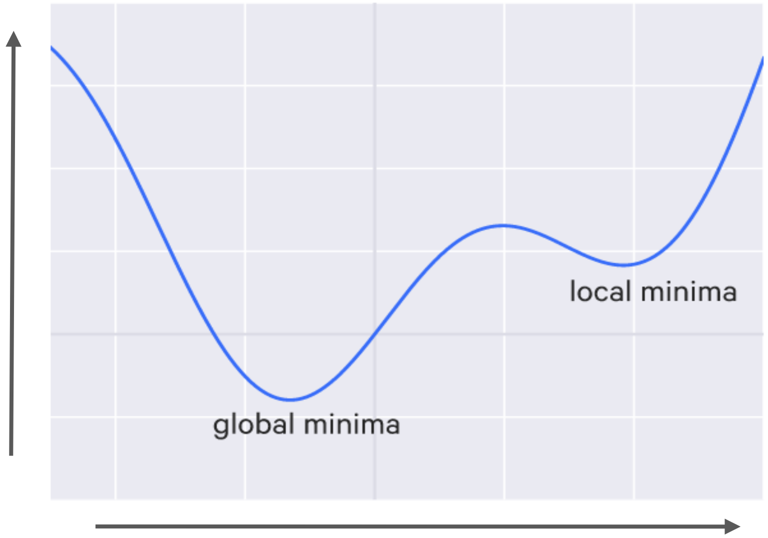

- But left very **general**, **not fleshed out**.

Novel contribution (G et al.)

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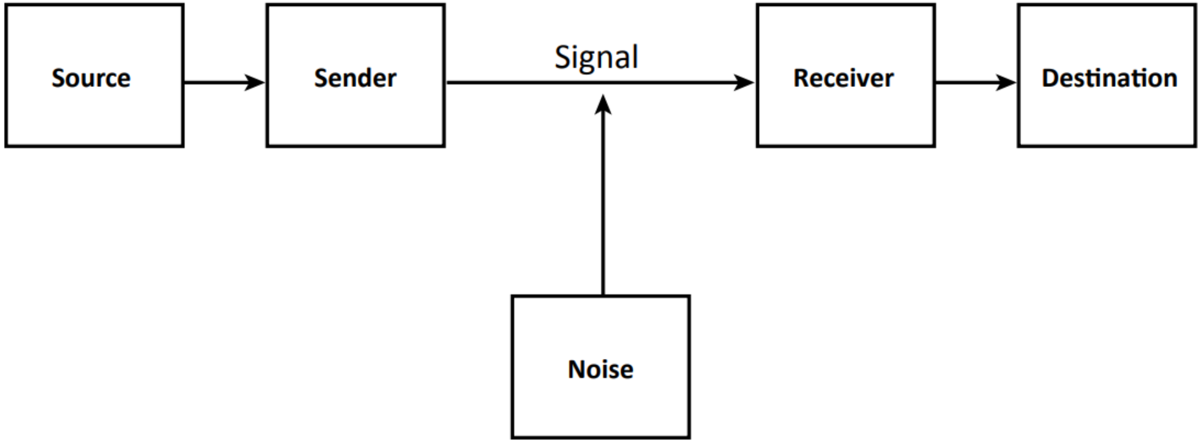
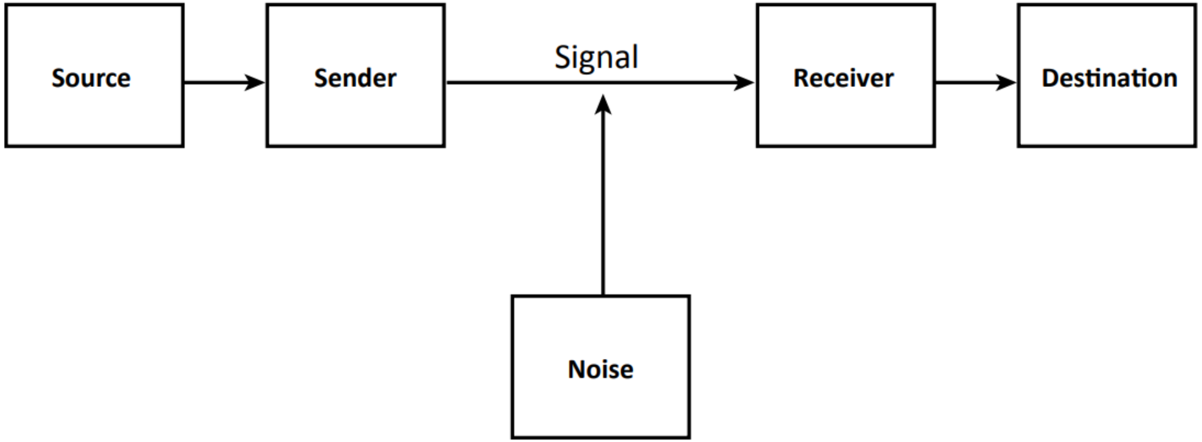
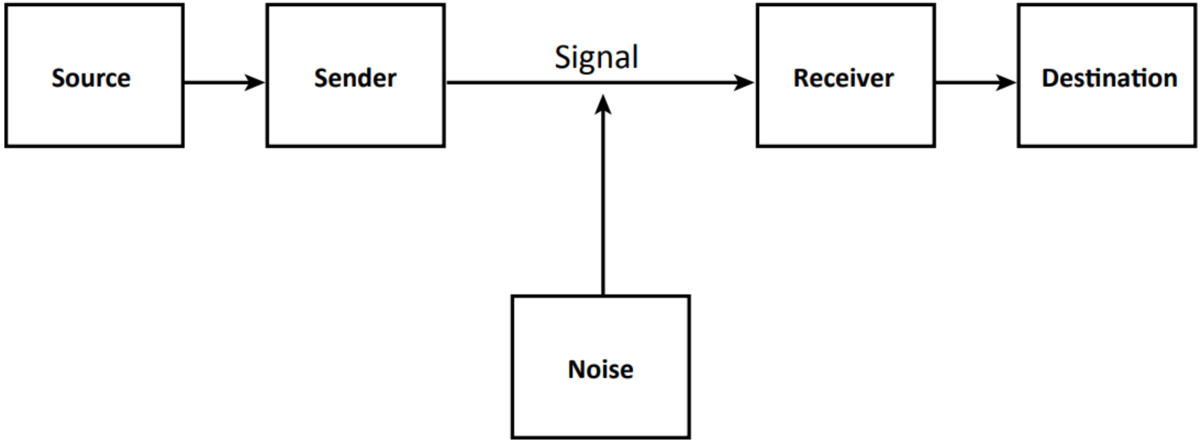
- The principle of **efficient communication**, often cast in terms of **information theory**, fleshes out the picture of **optimization** over some landscape.


Source
Sender
Signal
Receiver
Destination
Noise



- Successful communication: message received at destination is **identical** to message at source.



- Successful communication: message received at destination is **identical** to message at source.
- Efficiency: **successful communication** with **minimal effort** (e.g. length) by sender, receiver.

IDK, LOL, IMHO, WRT, WOLOG…

Basis for contribution (G et al.)

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<u>Ambiguity</u>:

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<u>Ambiguity</u>:

- Not a problem for communication because **context** often disambiguates.
- So you can **recycle** short forms (e.g. *to*).
- Leaving out information that is supplied by context makes communication more **concise**.

Basis for contribution (G et al.)

Word length:

- More **frequent** words tends to be shorter: e.g. ‘the’ versus ‘accordion’. (Zipf)

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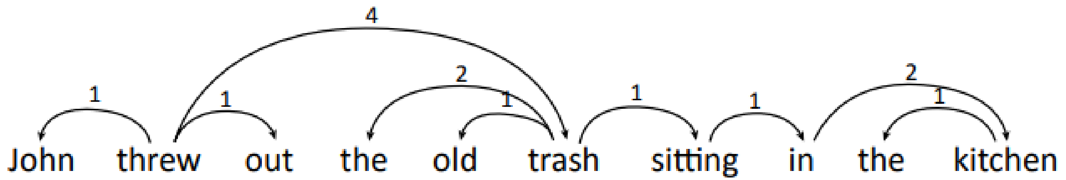
- Core notion is **predictability**, which can be affected both by frequency and by context.
- E.g. ‘chimp’/’chimpanzee’, ‘info’/’information’: shorter form more likely in predictive context.

Basis for contribution (G et al.)

Dependency locality in syntax:



Sentence 3 Total dependency length = 14

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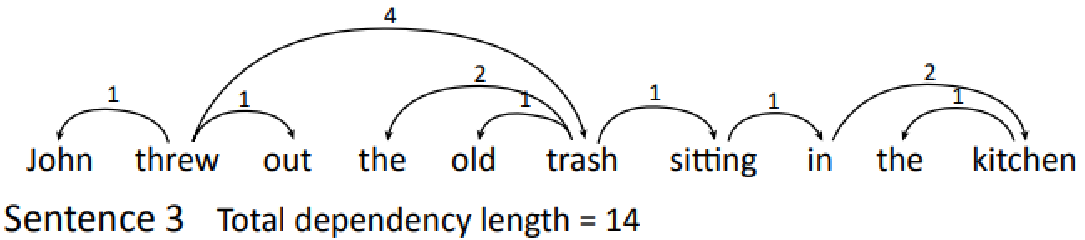

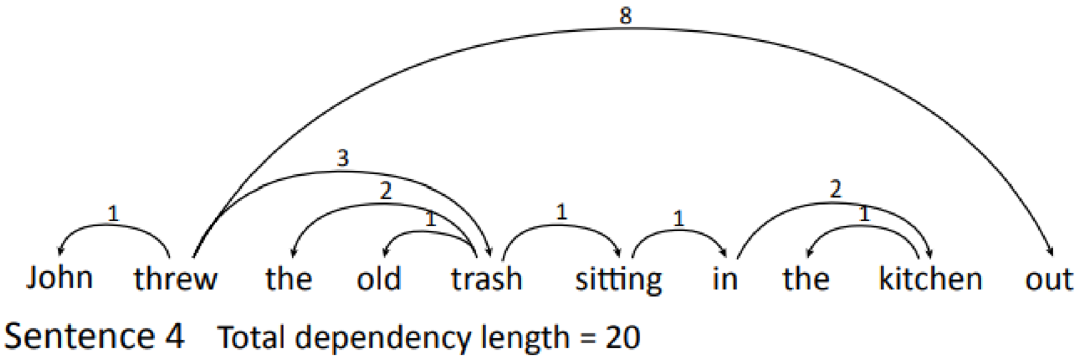

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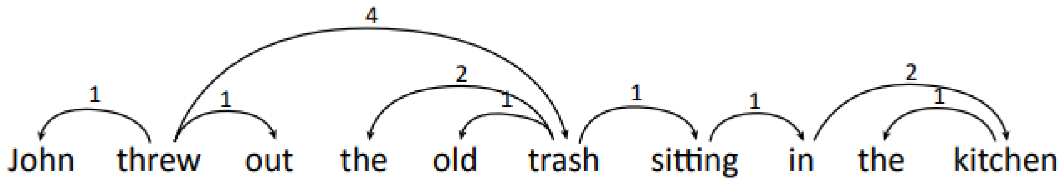
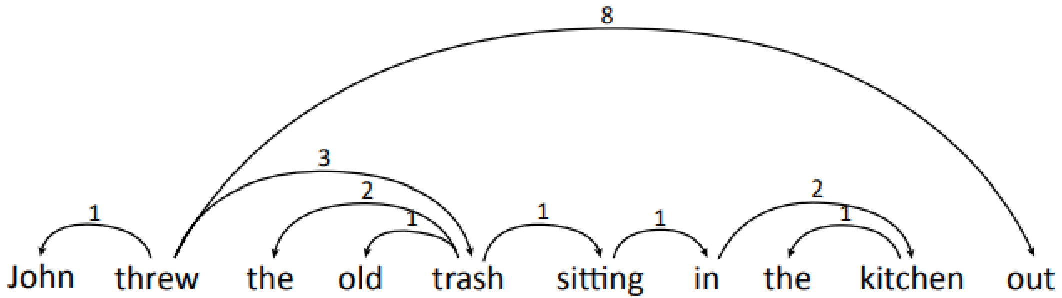


Sentence 3 Total dependency length = 14



Sentence 4 Total dependency length = 20

More effort!

Basis for contribution (G et al.)

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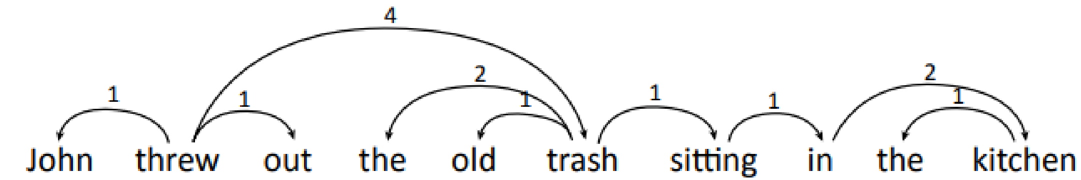
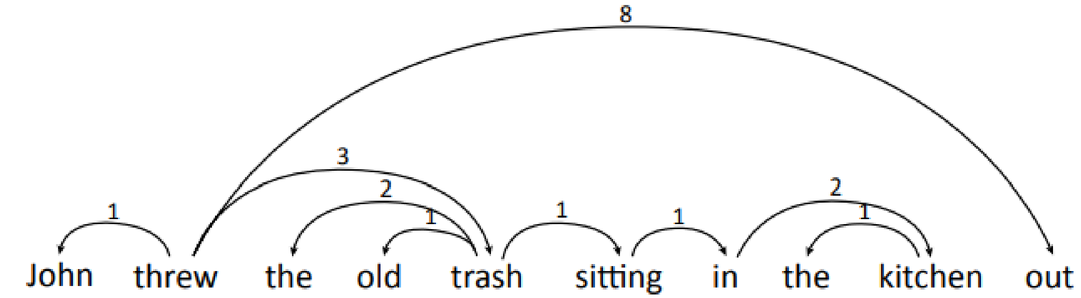


Sentence 3 Total dependency length = 14

Languages tend to minimize DL, and thus effort.

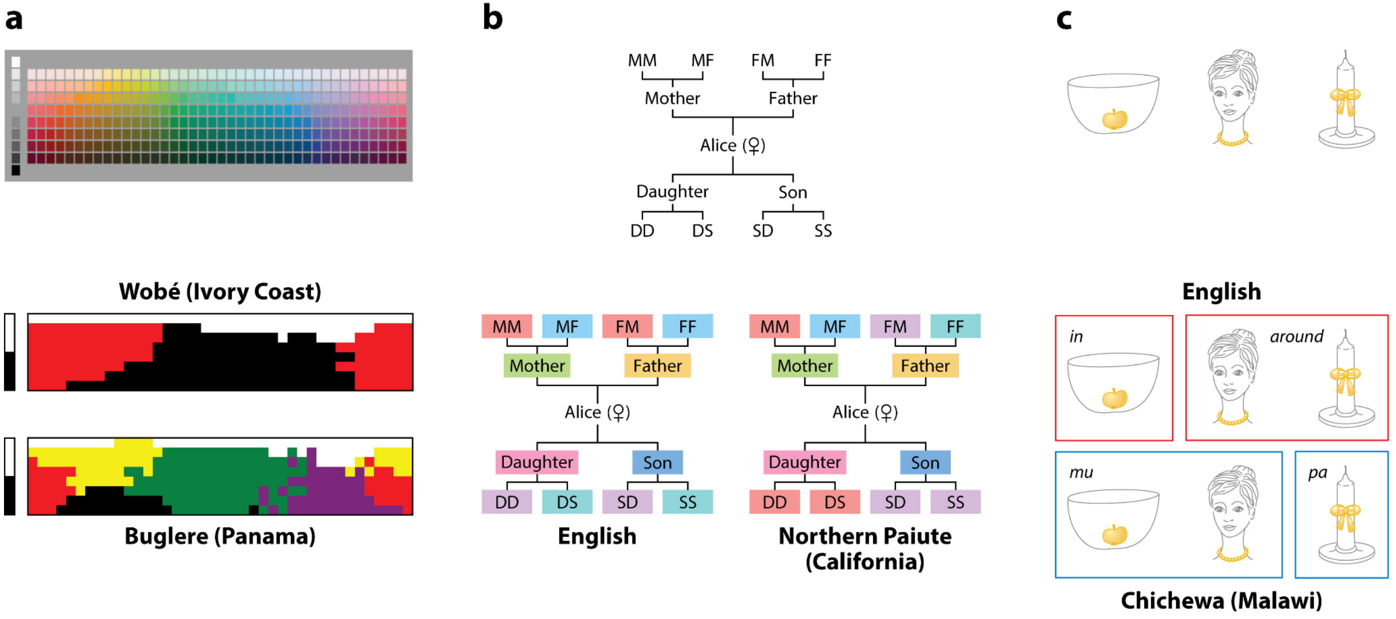


Sentence 4 Total dependency length = 20

More effort!

Basis for contribution (G et al.)

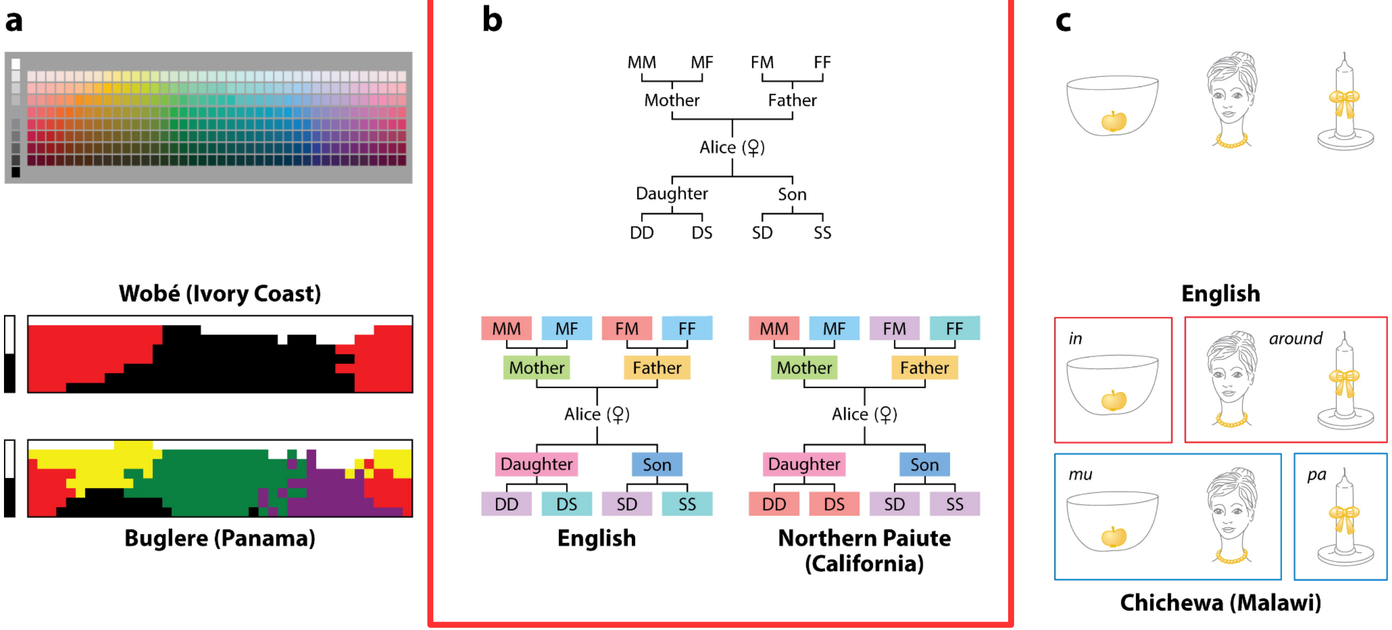
Semantic categories in the lexicon



Kemp C, et al. 2018.
Annu. Rev. Linguist. 4:109–28

Basis for contribution (G et al.)

Semantic categories in the lexicon



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Kinship Categories Across Languages Reflect General Communicative Principles

CHARLES KEMP AND TERRY REGIER Authors Info & Affiliations

SCIENCE · 25 May 2012 · Vol 336, Issue 6084 · pp. 1049-1054 · DOI: 10.1126/science.1218811

Charles Kemp

Background (K&R)

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- **Kinship** categories **vary** across languages but also show **universal** tendencies – that is, they exhibit **constrained variation**.

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- **Kinship** categories **vary** across languages but also show **universal** tendencies – that is, they exhibit **constrained variation**.
- Previous accounts have explained this pattern in terms of constraints that are mostly **specific to kinship** and not derived from **general principles** (although general considerations e.g. usage frequency are sometimes invoked).

Novel contribution (K&R)

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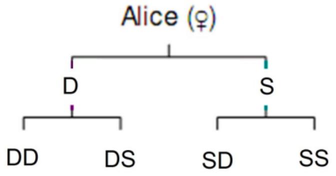
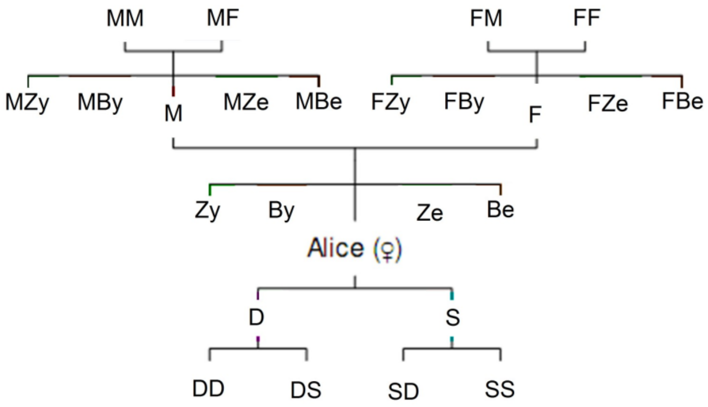
Novel contribution (K&R)

- Attested kinship systems achieve a **near-optimal tradeoff** between **informativeness** and **simplicity**.
- They are **efficient** in this sense.
- Informativeness and simplicity are both **domain-general** notions, although they must be operationalized on a **domain-specific** basis.

Basis for contribution (K&R)

Alice (♀)


Alice (♀)
D
S
DD
DS
SD
SS


MM
MF
FM
FF
MZy
MBy
M
MZe
MBe
FZy
FBy
F
FZe
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Zy
By
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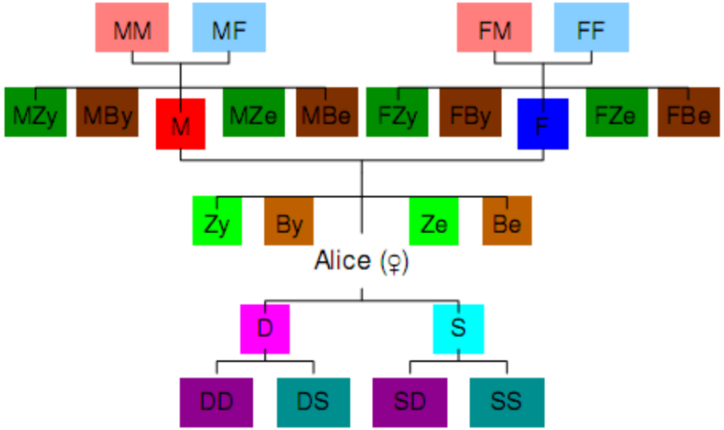

MM
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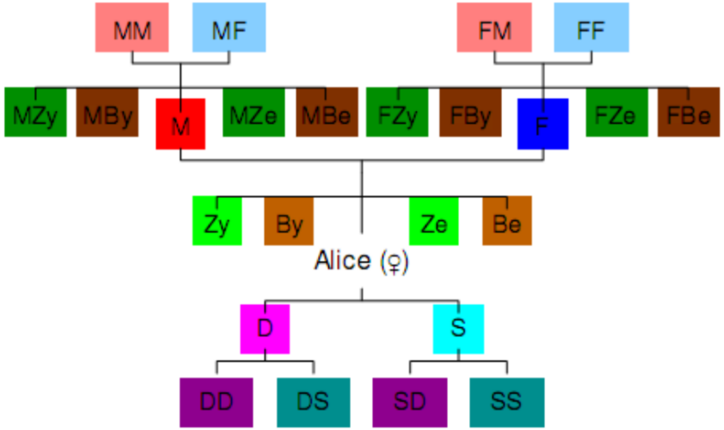
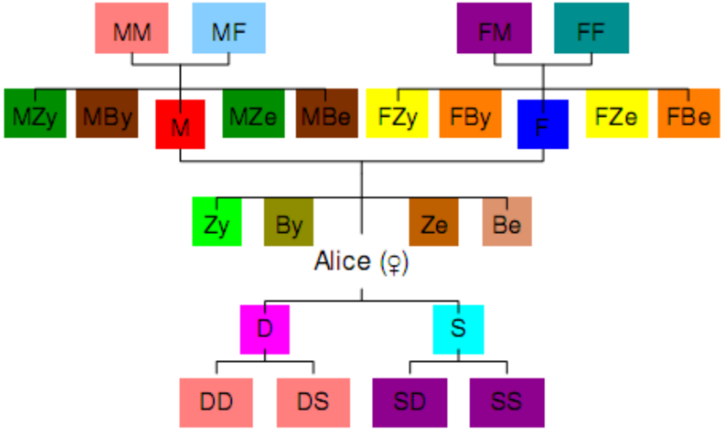
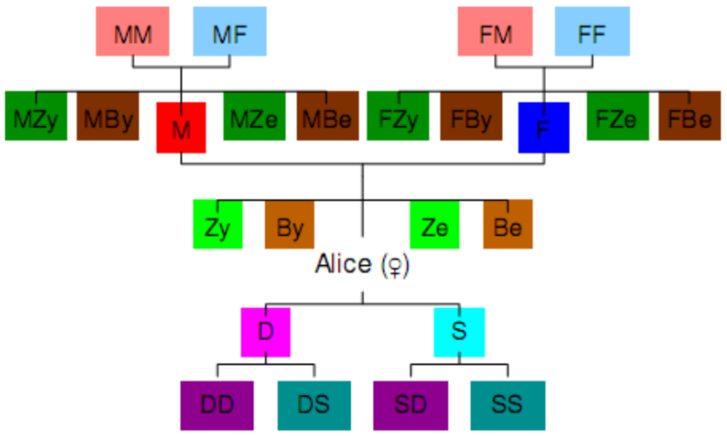
English



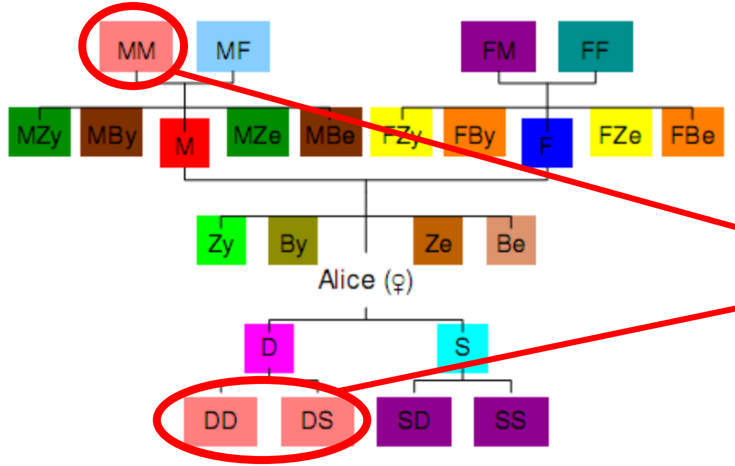
English



Northern Paiute

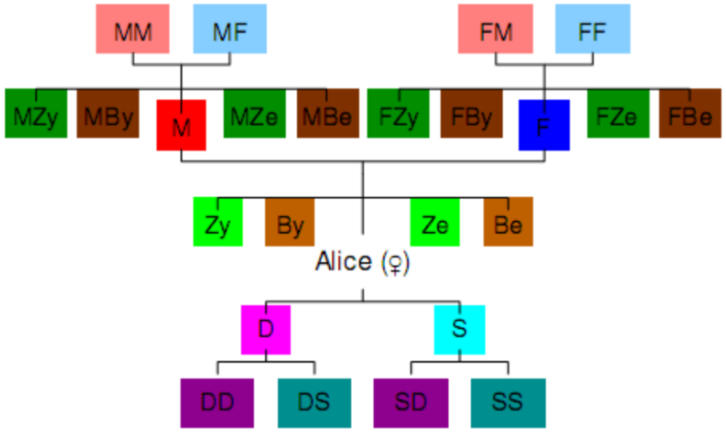


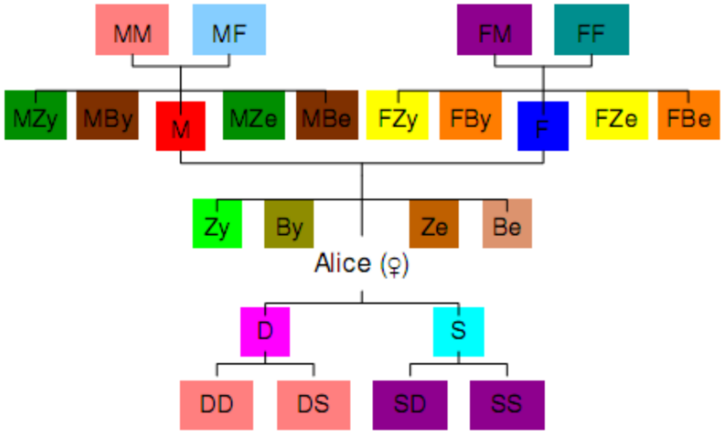
English



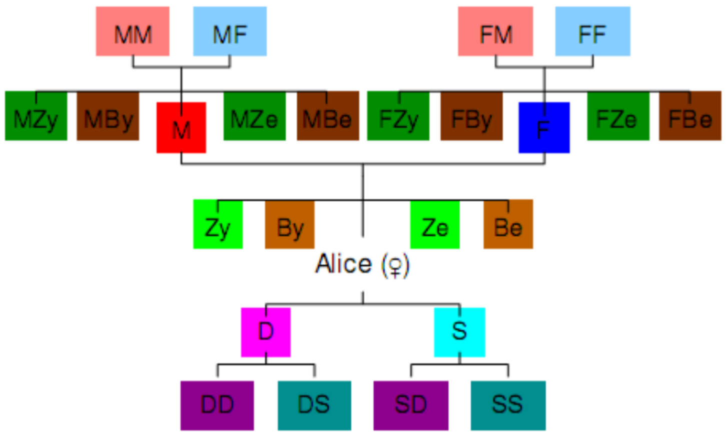
Northern Paiute

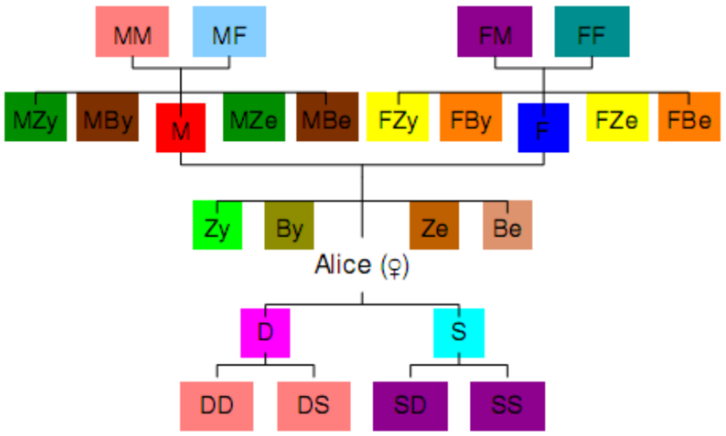
MM, DD, DS fall in the same category: **reciprocal MM**





Data: 487 languages (Murdock, 1970).





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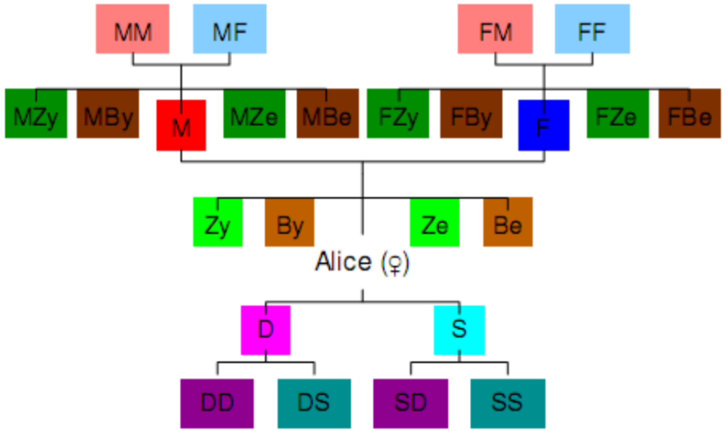
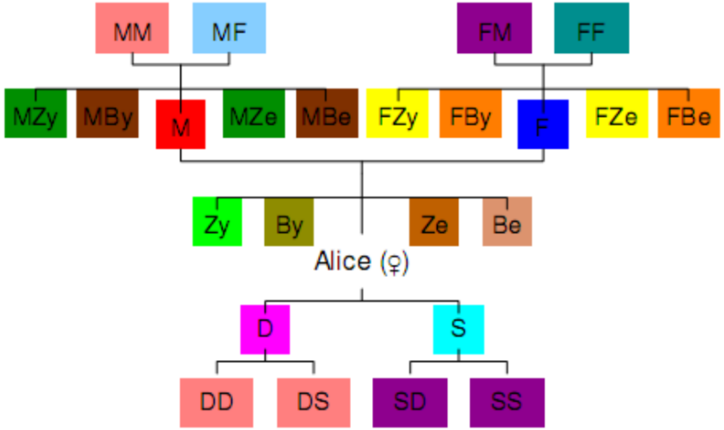
Representation: Kinship grammars, based on assumed universal primitives from literature.





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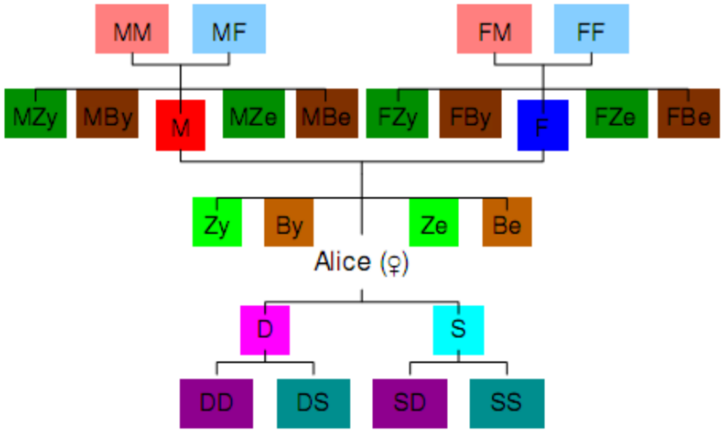

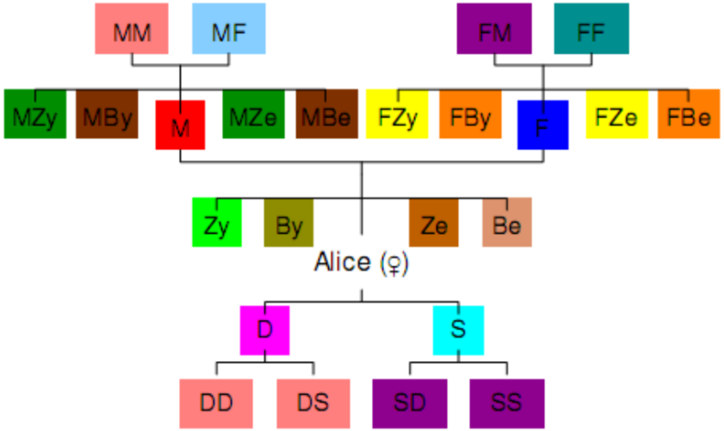

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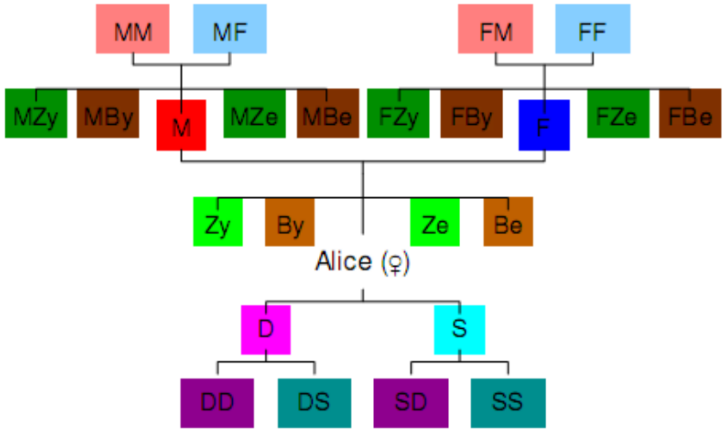

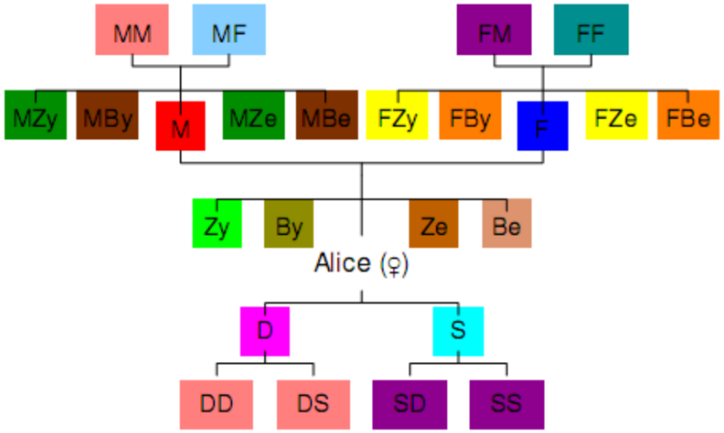

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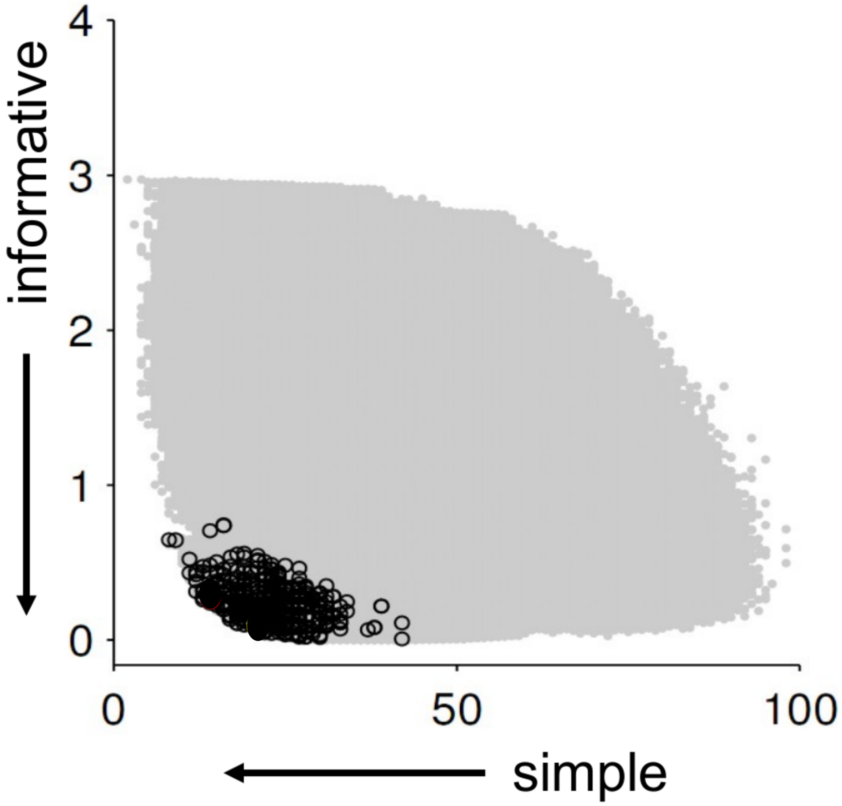

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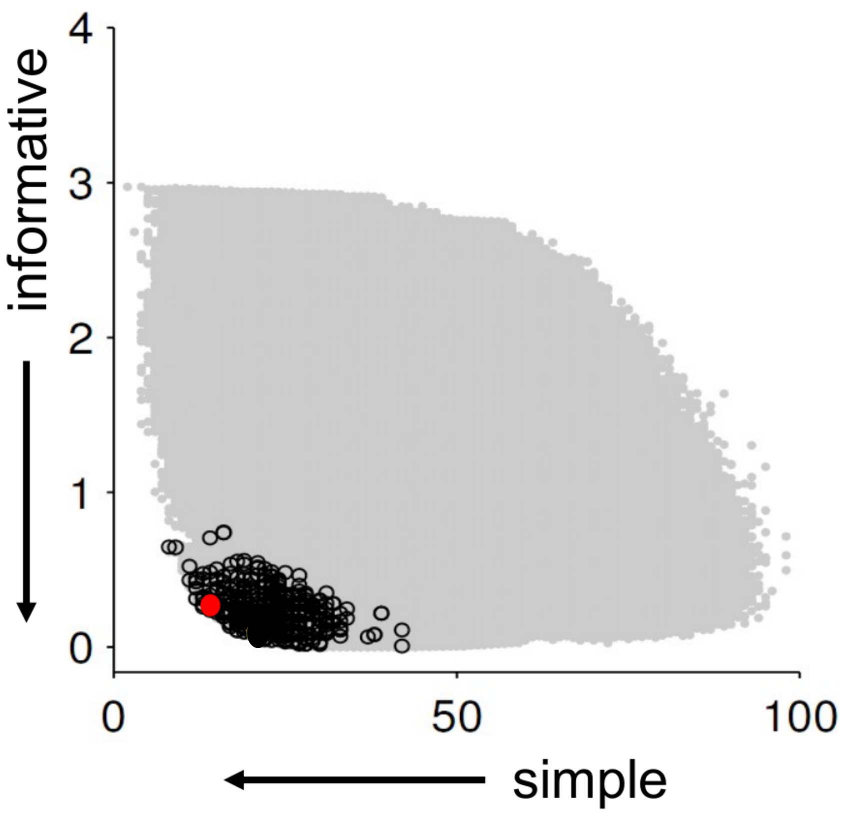

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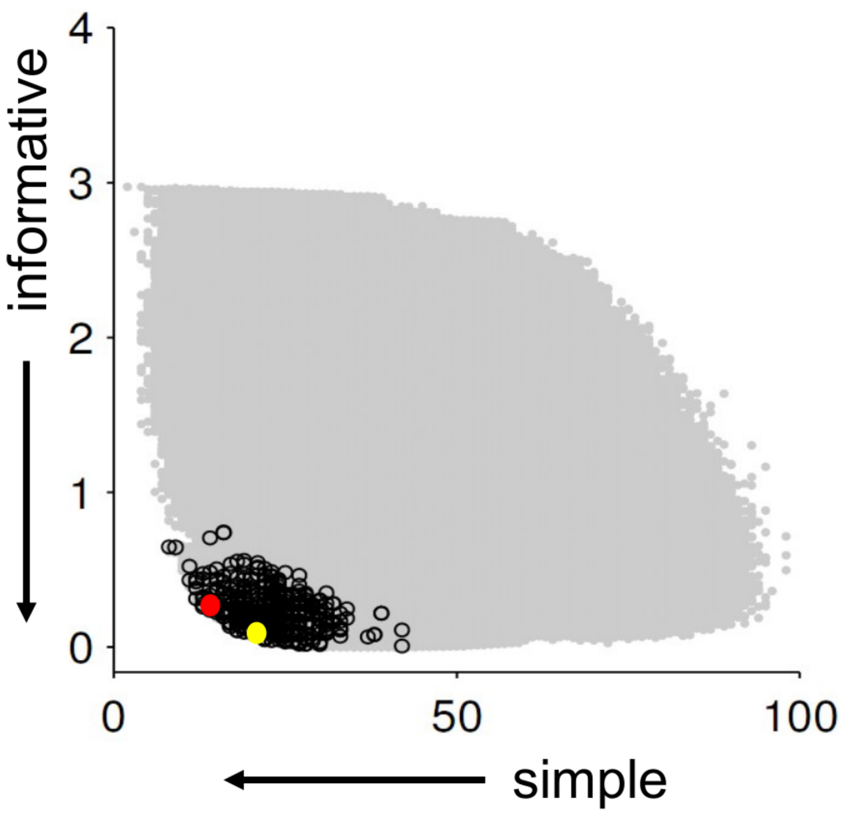

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How?

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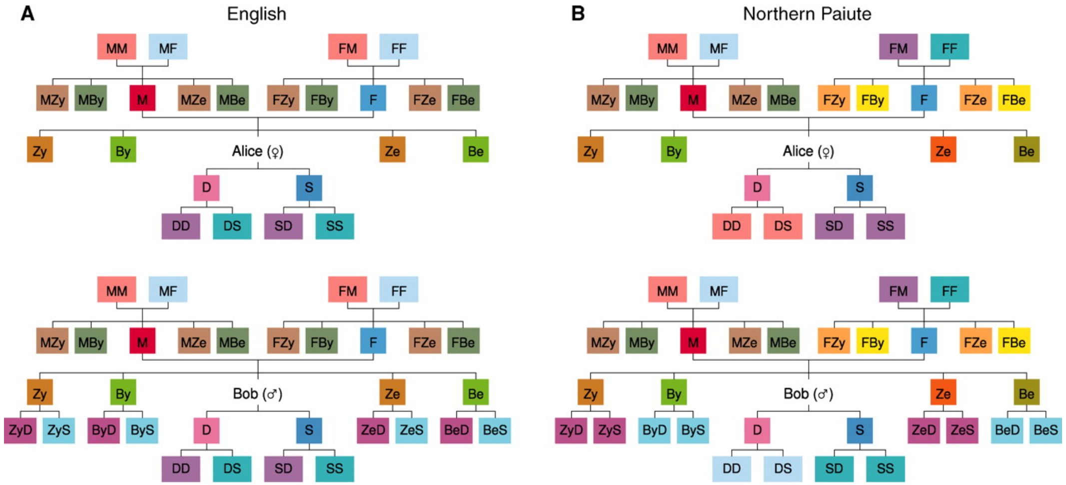
- Length is measured in **number of rules**.


A
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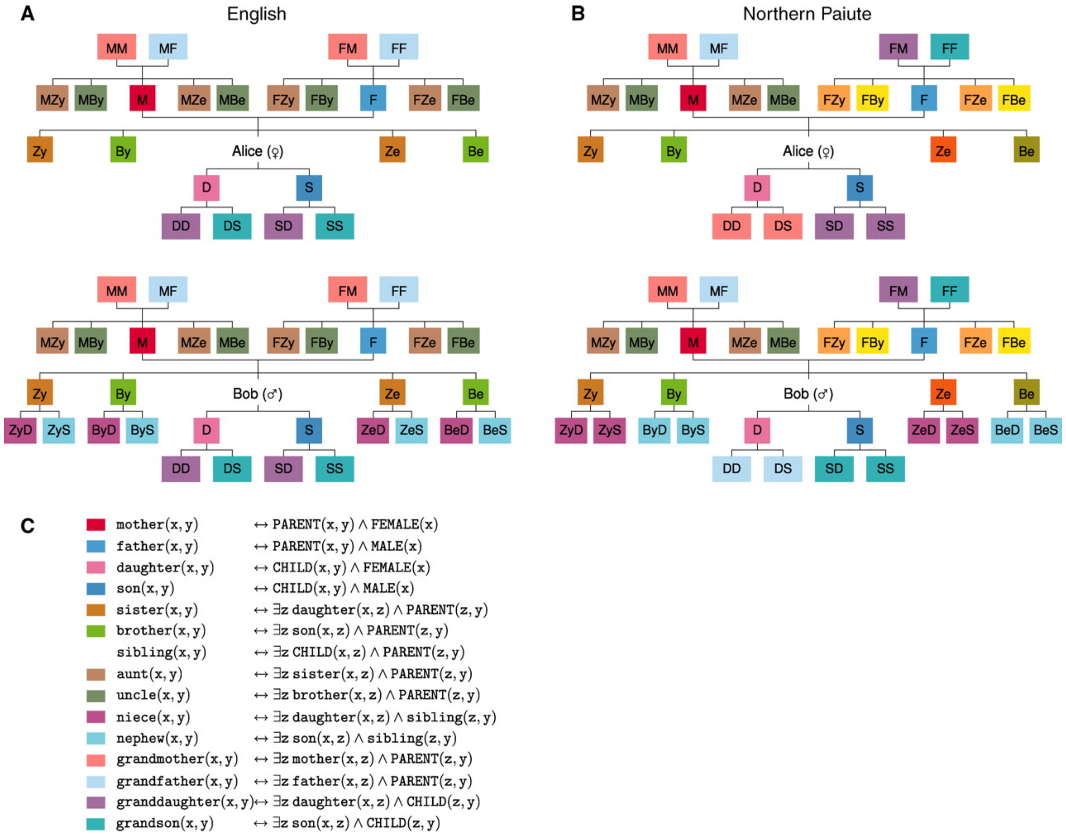
F
FZe
FBe
Zy
By
Alice (♀)
Ze
Be
D
S
DD
DS
SD
SS
Bob (♂)
ZyD
ZyS
ByD
ByS
ZeD
ZeS
BeD
BeS


A
English
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C
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father(x, y) ↔ PARENT(x, y) ∧ MALE(x)
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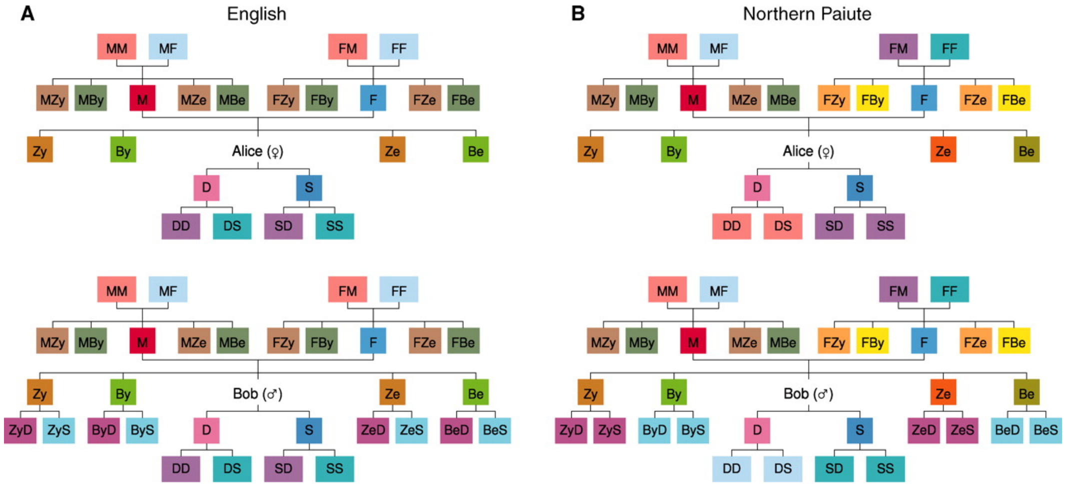

15 rules

C

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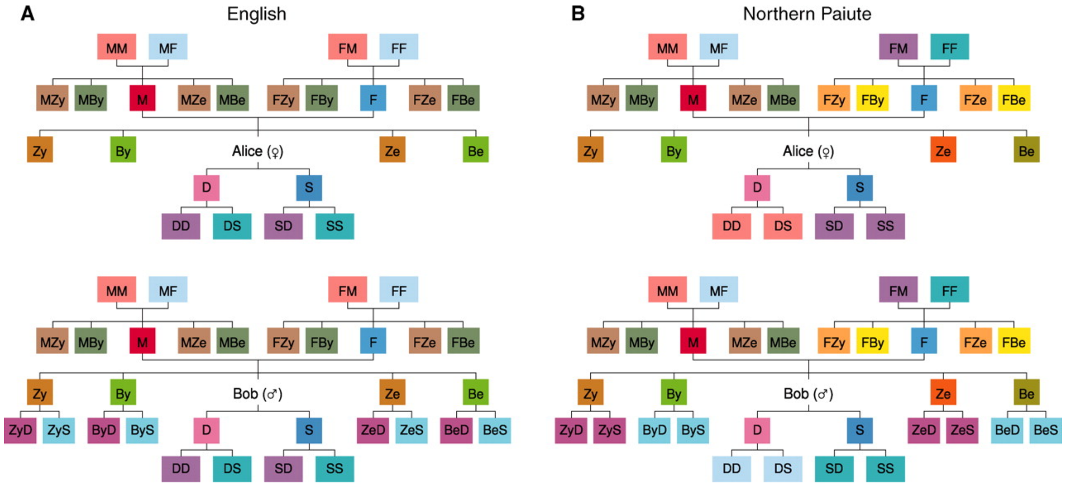

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D

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$\texttt{brother}(x,y) \leftrightarrow \exists z\, \texttt{son}(x,z) \wedge \texttt{PARENT}(z,y)$
$\texttt{youngersister}(x,y) \leftrightarrow \texttt{sister}(x,y) \wedge \texttt{YOUNGER}(x,y)$
$\texttt{oldersister}(x,y) \leftrightarrow \texttt{sister}(x,y) \wedge \texttt{OLDER}(x,y)$
$\texttt{youngerbrother}(x,y) \leftrightarrow \texttt{brother}(x,y) \wedge \texttt{YOUNGER}(x,y)$
$\texttt{olderbrother}(x,y) \leftrightarrow \texttt{brother}(x,y) \wedge \texttt{OLDER}(x,y)$
$\texttt{maternalaunt}(x,y) \leftrightarrow \exists z\, \texttt{sister}(x,z) \wedge \texttt{mother}(z,y)$
$\texttt{maternaluncle}(x,y) \leftrightarrow \exists z\, \texttt{brother}(x,z) \wedge \texttt{mother}(z,y)$
$\texttt{paternalaunt}(x,y) \leftrightarrow \exists z\, \texttt{sister}(x,z) \wedge \texttt{father}(z,y)$
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$\texttt{mansisterchild}(x,y) \leftrightarrow \texttt{maternaluncle}(y,x)$
$\texttt{manbrotherchild}(x,y) \leftrightarrow \texttt{paternaluncle}(y,x)$
$\texttt{maternalgrandmother}(x,y) \leftrightarrow \exists z\, \texttt{mother}(x,z) \wedge \texttt{mother}(z,y)$
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$\texttt{selfreciprocal1}(x,y) \leftrightarrow \texttt{maternalgrandmother}^{\leftrightarrow}(x,y)$
$\texttt{selfreciprocal2}(x,y) \leftrightarrow \texttt{maternalgrandfather}^{\leftrightarrow}(x,y)$
$\texttt{selfreciprocal3}(x,y) \leftrightarrow \texttt{paternalgrandmother}^{\leftrightarrow}(x,y)$
$\texttt{selfreciprocal4}(x,y) \leftrightarrow \texttt{paternalgrandfather}^{\leftrightarrow}(x,y)$

24 rules

Informativeness

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- Intuitively, a precise and narrow category like **“mother’s younger sister”** (MZy) is **more informative** than a broad category like **“aunt”** (MZy, MZe, FZy, FZe).

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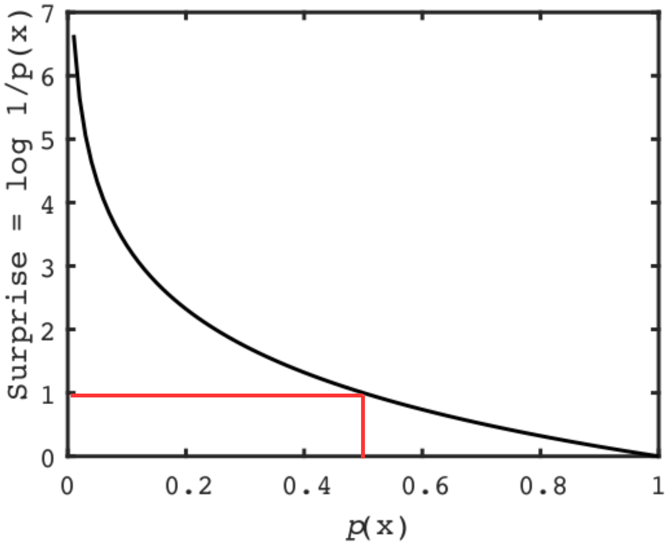
- Some **information is lost** when you use a word like “aunt”.
- **How much?**

Information as surprise. The Shannon information, or **surprisal**, of a particular outcome x_h (e.g. heads) is:

$$\log 1 / p(x_h) \text{ bits} \quad = \quad - \log p(x_h) \text{ bits}$$



p_i

Need probability: **how often** you will need to refer to this kin type

$$p_i$$

<u>Need probability</u>: **how often** you will need to refer to this kin type

$$c_i = -\log_2\left(\frac{p_i}{\sum_{z_j = z_i} p_j}\right)$$

<u>Communicative cost</u>: **how surprising** is it that the kin type is *i*, **given** that the category is *z*?

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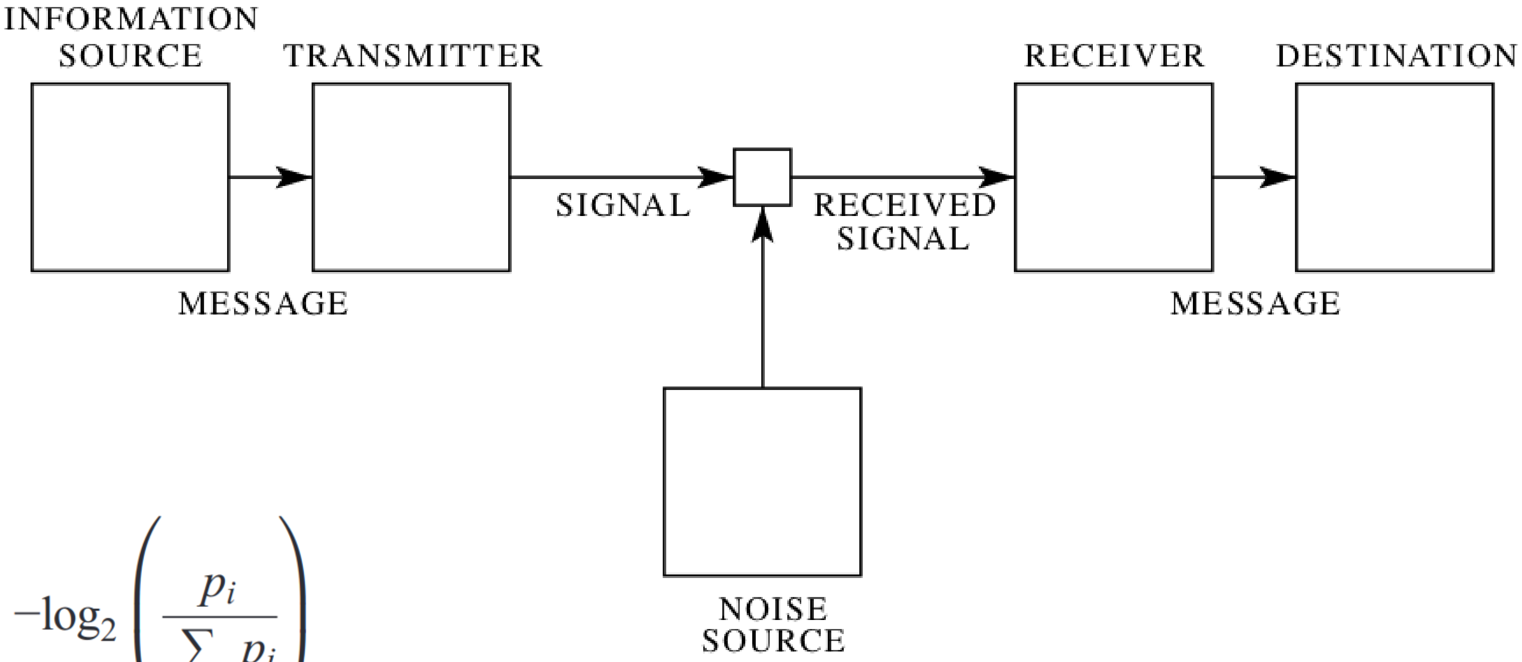

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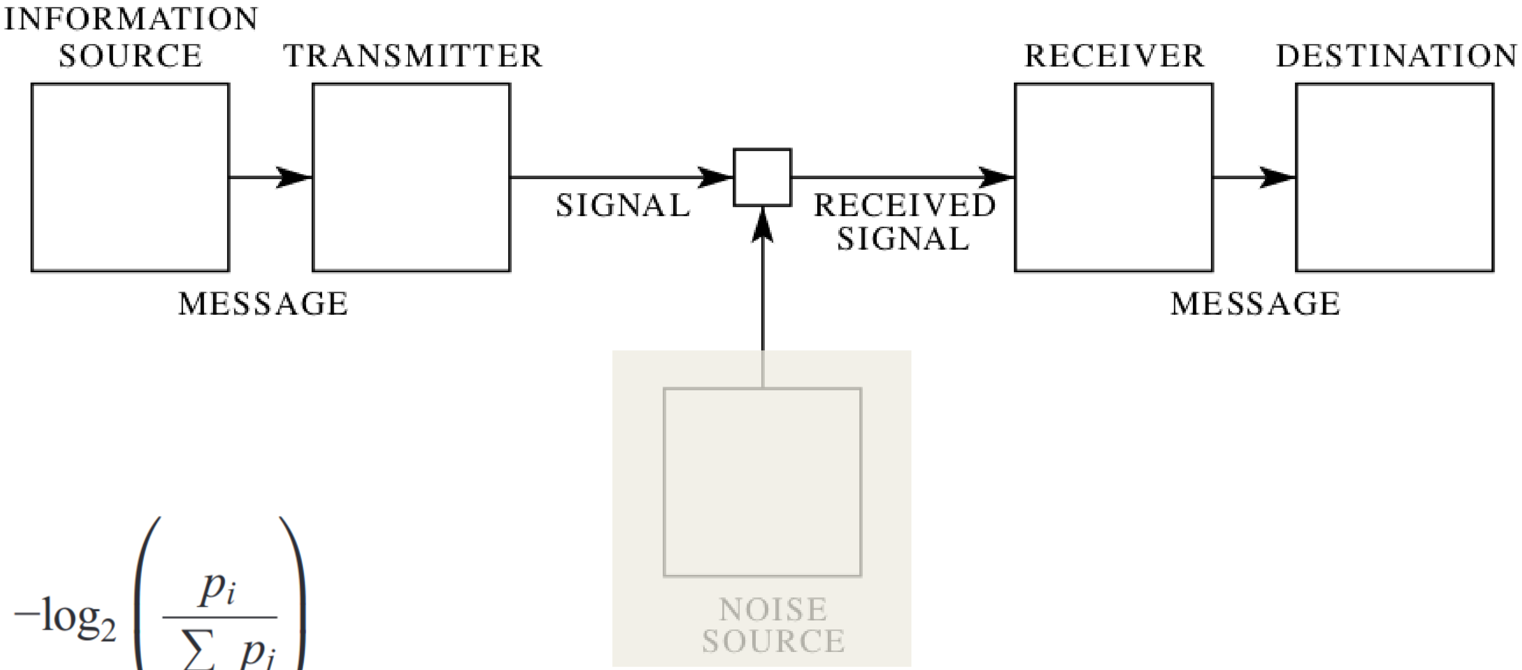

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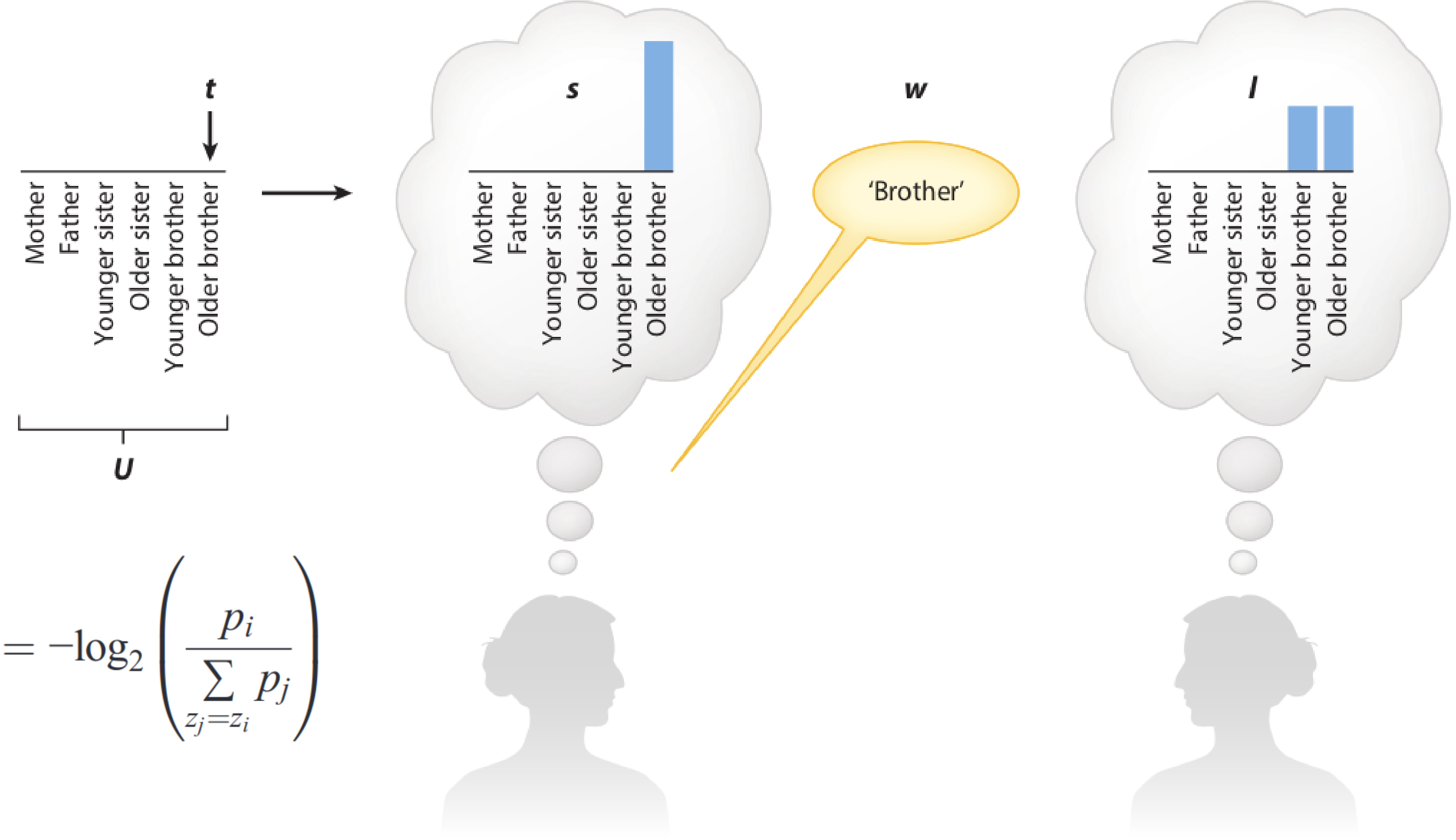

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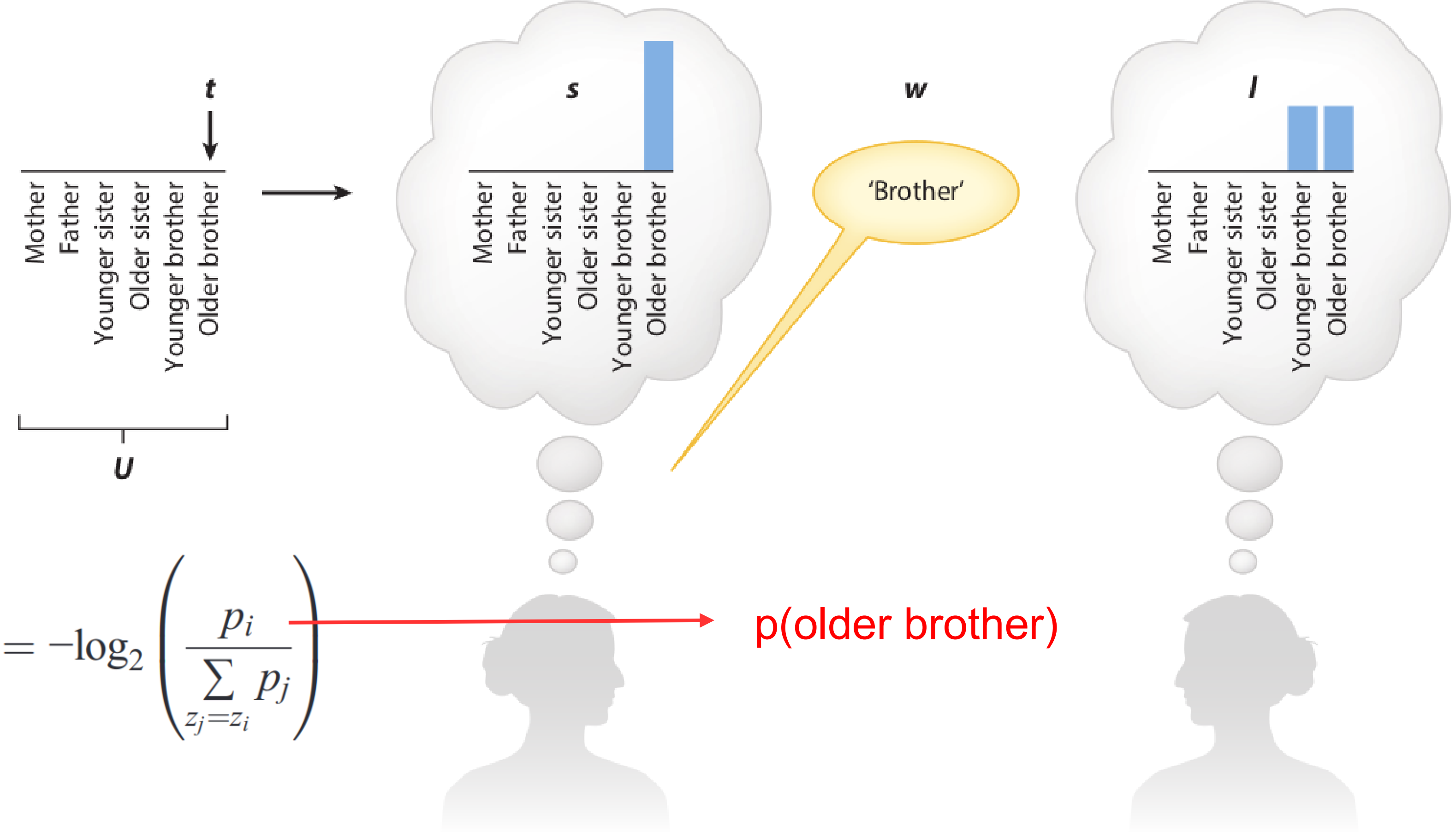

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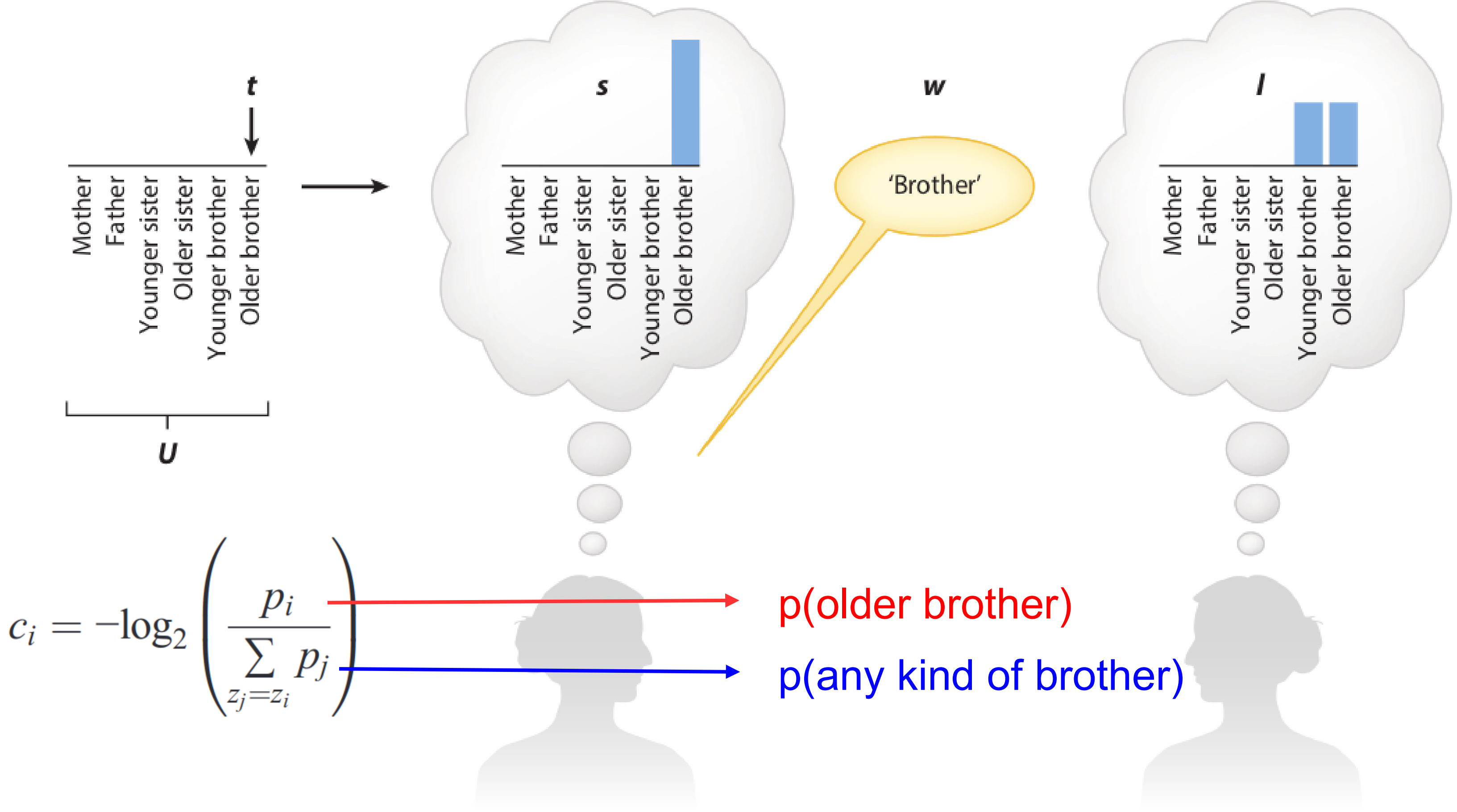

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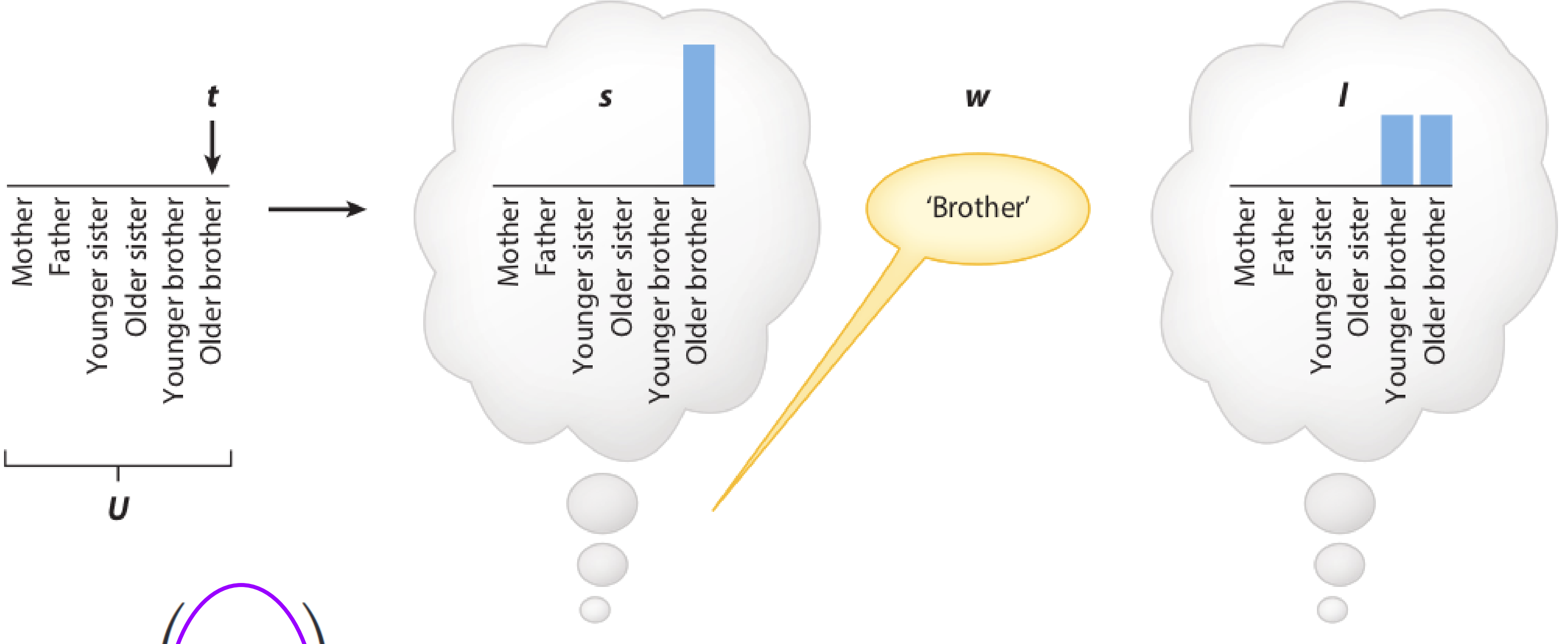

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p(older brother)

p(any kind of brother)

p(older brother | some kind of brother)

$$p_i$$

<u>Need probability</u>: **how often** you will need to refer to this kin type

$$c_i = -\log_2\left(\frac{p_i}{\sum_{z_j=z_i} p_j}\right)$$

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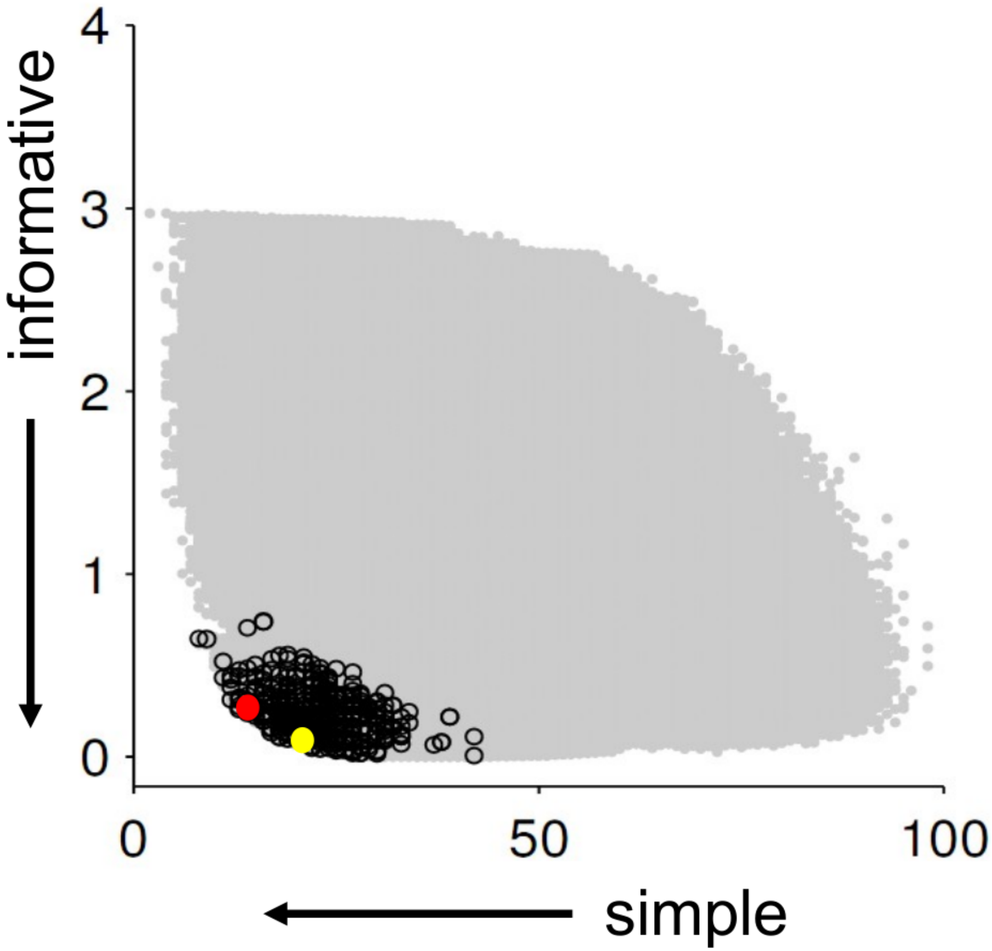
$$C = \sum_{i=1}^{24} p_i c_i$$

<u>Overall communicative cost</u>: what is the **expected communicative cost** over the entire kinship terminology?



informative
4
3
2
1
0
0
50
100
simple

Comments?
Questions?

Explaining known constraints

More semantic distinctions in:

1. Near relatives (e.g. siblings), relative to distant relatives (e.g. maternal siblings)

Greenberg 1966

Explaining known constraints

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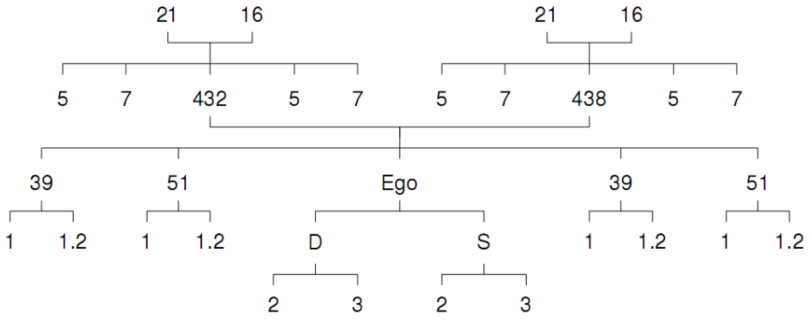

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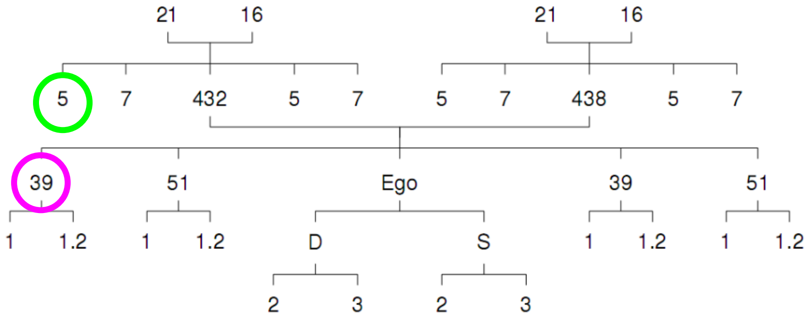

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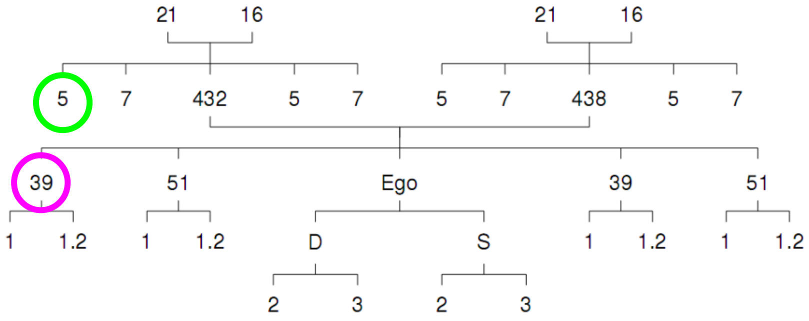

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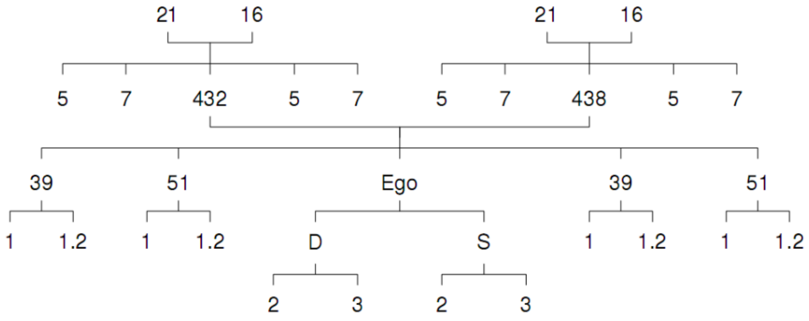

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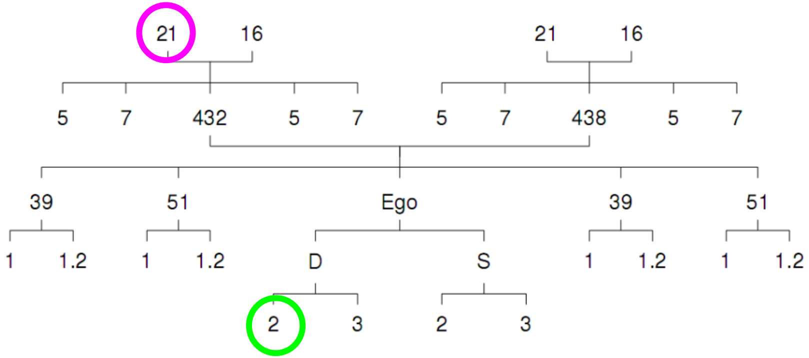

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Intended takeaway message

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- Attested kinship systems achieve a **near-optimal tradeoff** between **informativeness** and **simplicity**.

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- Attested kinship systems achieve a **near-optimal tradeoff** between **informativeness** and **simplicity**.
- They are **efficient** in this sense.

What **questions** do you have?

What **comments** do you have?